

GE Healthcare

Signa HD to HDxt System Upgrade for 1.5T and 3.0T

Technical Publication



5166004
Rev 7

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
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Table of Contents

DESCRIPTION	PAGE
Table Of Contents	1
Language Policy For Service Documentation	3
Revision History	5
SECTION 1 – Getting Started	7
SECTION 2 – Upgrade Installation Process	12
SECTION 3 – Power Off	16
SECTION 4 – System Upgrade Cables	17
SECTION 5 – Operator Workspace Upgrade	27
SECTION 6 – System Cabinet Upgrade	49
SECTION 7 – Completion	75

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
2-14-2006	1	Initial Release
6-26-2006	2	Section 2-3: Added CAUTION note to upgrade flowchart that refers to the IIS Key and SaveINFO operation. Revised RFS System cabinet flow. Revised s/w load instructions and added LCD monitor adjustments (MRIhc15805, MRIhc16251, MRIhc16325). Added blocks for restoreINFO, ICW, and update of eLicense (MRIh15873, MRIhc5892, MRIhc5894, MRIhc6209). Section 4-3: Revised the cable relabeling information by adding CAUTION Note and emphasizing the rerouting of cables to J5 and J10 on Driver Module. Section 4-4: Revised cable maps to reflect relabeled cable routing of Runs 1329 and 1330. Added existing term server cables connected to ports 3 and 4 (MRIhc5271). Switched ethernet cables 5145044-5 & 5145044-7 in System and RF cabinets (MRIhc5895). Section 5: Corrected info on Run 2053, orientation of Humidity iButton, and added more info for using only CAT6 cable, Run1323. (MRIhc16091). Added note informing the installer not to connect additional monitors to host computer prior to operating system software load (MRIhc15870). Section 6: Added note to install ICN modules in correct orientation (MRIhc16057). Added note not to disconnect cables that are not part of this upgrade (MRIhc16127).
2-19-2007	3	Section 2-3: Reversed flowchart procedures <i>Software Option Install w/eLicense</i> and <i>Install and Calibration Wizard</i> . Section 5: Added note that Humidity iButton can be installed in either direction. Section 6: Updated instructions for installing new Power Panel in 1.5T FRS cabinet. Revised installation location of ICN brackets (CQA 13078452). Added relabeling information for cables connected to Driver Module (CQA 13091096).
5-23-2007	4	Section 4: Clarified USB-Serial Adapter/Serial-1Wire Adapter in cable maps. Section 5: Updated iButton positioning. Added block diagram for Environmental Sensor installation.
6-12-2008	5	Section 2: Added note for installing IIQ "S" bracket, if required (PQR 13162911). Section 4: Revised cable maps to show upgrade of GOC components and cables, not the complete GOC as was previously replaced. Updated the temperature/humidity sensor installation to new design. Section 5: Revised upgrade instructions for replacing components in GOC rather than the entire GOC cabinet (PCN 290212). Revised the temperature/humidity sensor installation procedure. Revised Barcode upgrade instructions (PQR 13121131).
9-18-2008	6	Updated information in manual to include HDxt (Release 15.x) where applicable. Section 6: Revised installation of VRE modules with new SUN ICN's (ECO 2062202).
9-27-2010	7	Section 2: Updated eLicense model type to MR from Signa HDx (per Direction 2397845). Section 4: Updated cable maps for removal of OW components and installation of Z400 GOC (ECO 2096124). Section 5: Updated instructions for removal of obsolete components and GOC. Added instructions for installing Z400 GOC and new components (ECO 2096124). Section 6: Removed second set of ICN brackets in cabinet because new Sun 4170 ICN replaces two ICN's installed in previous upgrades (ECO 2101939).

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	7								
1 to 76	7								

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SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 1.5T or 3.0T HD (Release 12.x) to Signa HDx (Release 14.x) or HDxt (Release 15.x). Refer to illustration 1-1 for pictorial overview of areas of system upgraded.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 3.0T Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the HDx upgrade to GoldSeal which includes:

- De-installed Modules
- Cables

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION

1-7-5-1 Sites in Americas

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997
Fax: (414)-747-6855
Field ASPEN: 1-800-525-1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41
Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page – http://supportcentral.ge.com/15563](http://supportcentral.ge.com/15563)

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

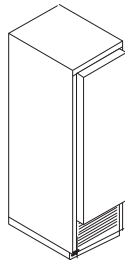
Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

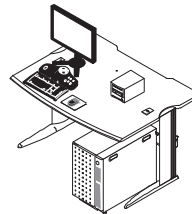
- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Confirm back-up site information. Shut site down.
- Remove obsolete equipment and cables.
- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-10 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment replaced and modified by Signa HD to HDx/HDxt Upgrade.



RFS or SYSTEM CABINET
(MR2)



OPERATOR WORKSPACE
(OW1)

SIGNA EXISTING HARDWARE REPLACED OR MODIFIED FOR UPGRADE TO HDx or HDxt
ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2-2 UPGRADE PROCEDURE

Note

For a system being upgraded, whether it is a fixed-site or a mobile, if the gradient lead restraint is using the 90 degree mobile van bracket or the IIQ “S” bracket assembly, do not modify or upgrade it. If the system being upgraded does not contain either of these, then install the IIQ “S” bracket assembly.

The following upgrade flow charts contain information for upgrading to Signa HDx 1.5T Release 14.x or HDxt 1.5T Release 15.x from Signa EXCITE HD Upgrade Release 12.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa HDx/HDxt equipment has been delivered together. **Make sure that every part required is available before starting.**

2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

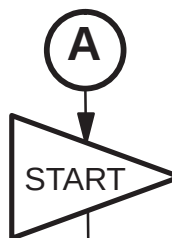
Procedures to be Completed by Field Engineer

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

CAUTION

Do not proceed without an IIS Key
on the SaveINFO media. Installation
calibration wizard will not permit the
exit from Installation mode in 14.x
without IIS Key.



Signa EXCITE HD to Signa HDx
1.5T or 3.0T UPGRADE CHECKLIST

☐ RE-GENERATE IIS KEY.
This must be done before upgrade starts to
make sure that sytem meets specifications.

☐ PERFORM "saveINFO" PROCEDURE
using a DVD

Procedures to be Completed by FE and/or Mechanical Installers

Section 1
☐ Section 1-5
PRODUCT DELIVERY INSTRUCTIONS

Section 1
☐ Section 1-6
DAMAGE IN TRANSPORTATION

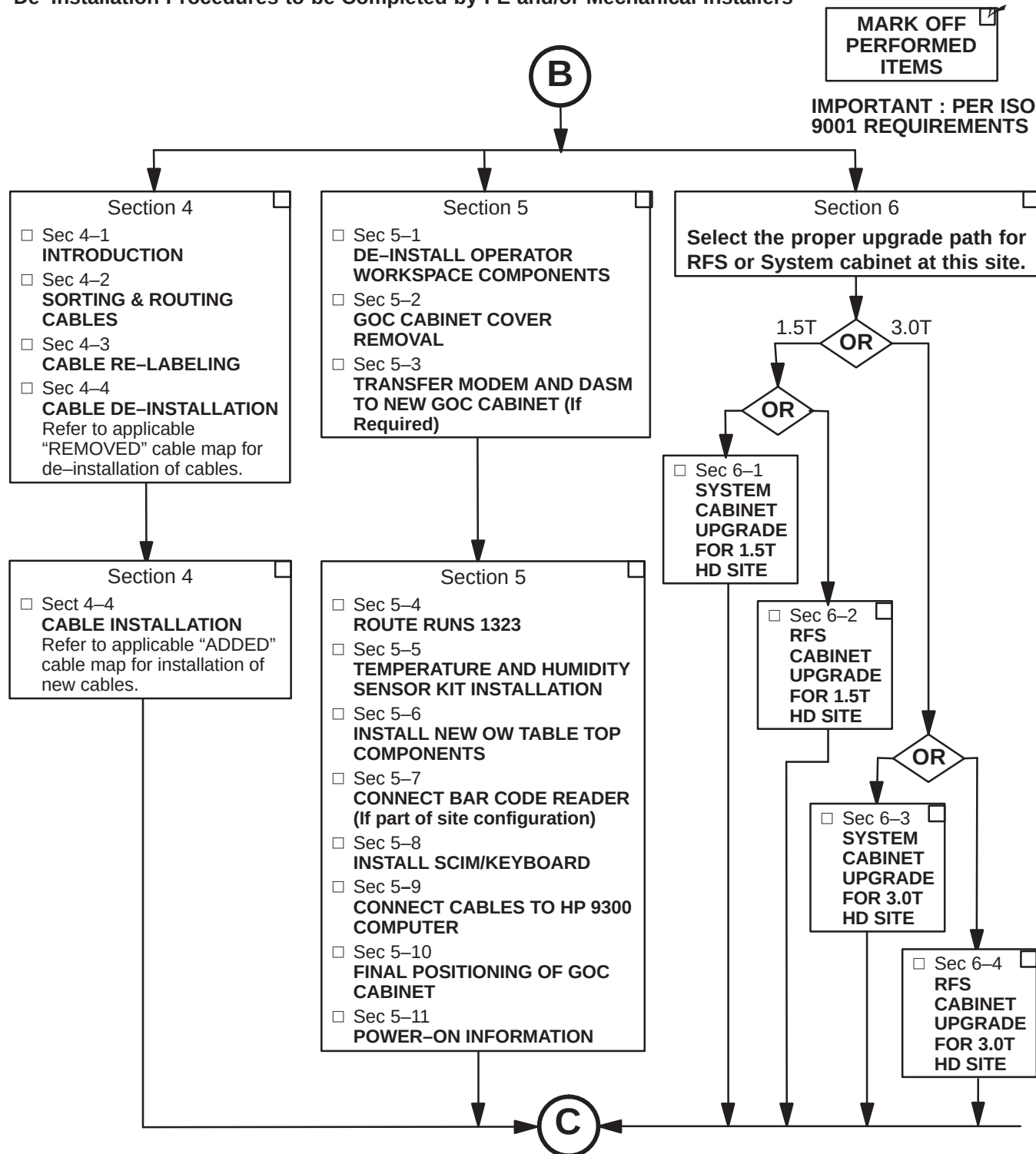
Section 1
☐ Section 1-8
PRODUCT LOCATOR

Section 3
☐ Section 3-1
SYSTEM POWER SHUTDOWN
☐ LOCKOUT & TAGOUT MAIN DISCONNECT



2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

De-Installation Procedures to be Completed by FE and/or Mechanical Installers



2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



IMPORTANT : PER ISO 9001 REQUIREMENTS

C

Section 8

- ☐ Section 7-1
MECHANICAL INSTALLATION COMPLETION CHECKLIST
- ☐ Section 7-2
RETURNING SHIPPING EQUIPMENT
- ☐ Section 7-3
MATERIAL DISPOSITION GUIDANCE
- ☐ Section 7-4
FINAL COORDINATION

MECHANICAL INSTALLATION COMPLETE

Procedures to be Completed by FE and Applications

MR Service Methods CD-ROM
Direction 5155500

- ☐ Install/Cals>System>Global Operator Console>
LCD Monitor Screen Adjustments

Section 7

- ☐ Section 7-5
POWER ON

If Host Software is NOT loaded?

If Host Software is loaded?

OR

MR Service Methods CD-ROM
Direction 5155500

- ☐ Software>System Host Software>
Host Operating System Software Load

MR Service Methods CD-ROM
Direction 5155500

- ☐ PERFORM "restoreINFO" PROCEDURE
Software>System Host Software>
MR Apps Software Load
STEPS 16 TO 20

MR Service Methods CD-ROM
Direction 5155500

- ☐ Installation>Options>
Software Option Install with eLicense
When performing this procedure the following must change:
HOST ID (See eLicense manual)
MODEL TYPE (change to MR)
- ☐ Installation>System>
Install and Calibration Wizard
- ☐ Installation>System>
Install and Calibration Wizard
– Start and then exit Grafidy to enable Grafidy files.

Applications/Training

- ☐ **Turn site over to Applications who will instruct users.**

TURN OVER TO CUSTOMER

Note:
HD to HDx upgrade does not require Grafidy calibration.

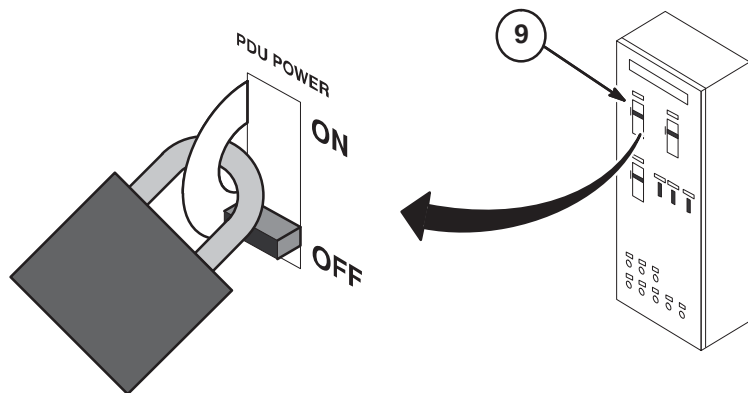
SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
2. Make sure that site customer has archived all patient information. Execute “Save Info” using a DVD.
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Perform normal software shutdown procedures.
6. Press the POWER OFF button on the PDU front control panel.
7. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
8. Locate and shut off main disconnect supplying power to the PDU.
9. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
10. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring



SECTION 4 – SYSTEM UPGRADE CABLES

4-1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading a Signa HD (Release 12.x) to a Signa HDx (Release 14.x) or HDxt (Release 15.x) for 1.5T and 3.0T.

- The “Affected” (removed) Cable maps are located on pages 19 thru 22.
- The “Affected” (added) Cable maps are located on pages 23 thru 26.
- Depending on site requirements, certain upgrades will require the installation of a complete new GOC with Z400 computer and new GOC power cord

Tape map in a convenient location to check off routed cable. Upon completion of connecting cables, return fold-out map to the binder for future reference.

4-2 SORTING AND ROUTING CABLES



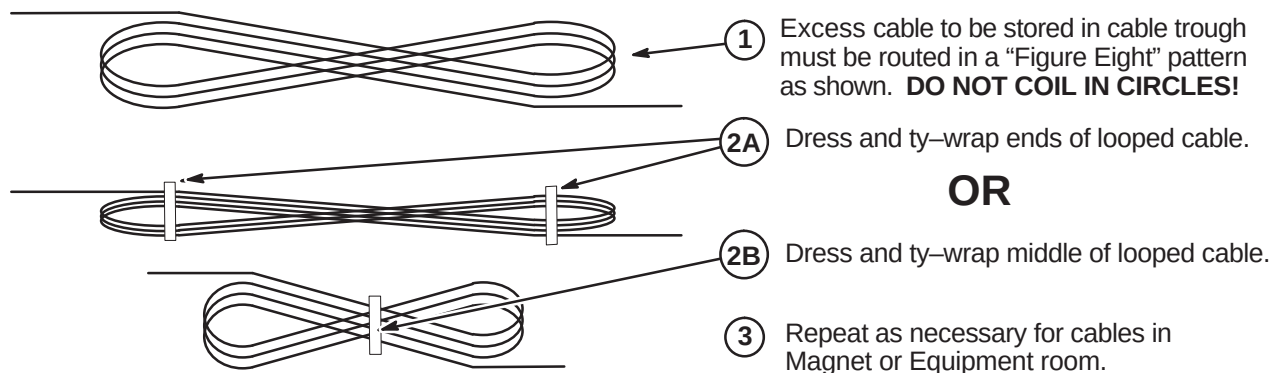
Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 4-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4-3 CABLE RE-LABELING

Relabel existing cables as described below:

- The new GOC upgrades the WIM to a GOCAA. This requires that Run 1084 be relabeled as Run 1255. Attach blank label to each end of cable and mark with information below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1084 MR1-A11-J26 2354601	RUN #1255 MR2-A11-J26 5120763-3	RUN #1084 OW1-J21-J6 VIA RUN #1062 2354601	RUN #1255 OW1-A21-J18 5120763-3



One of the following relabeling actions must be completed for this upgrade. Failure to do so will cause MC Bias Faults.

- For systems that were upgraded to HD prior to this HDx upgrade – Swap and relabel existing equipment room cable Runs 1030 and 1251 as instructed below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1030 PP1-J78 2329624	RUN #1329 PP1-J78 5160765	RUN #1030 MR2-A11-J61 2329624	RUN #1329 MR2-DVR-J10 5160765
RUN #1251 PP1-J203 5113186-11	RUN #1330 PP1-J203 5160765-4	RUN #1251 MR2-J66 5113186-11	RUN #1330 MR2-DVR-J5 5160765-4

- For systems that were installed new at HD (with RFS cabinet) prior to this HDx upgrade – Swap and relabel existing equipment room cable Runs 1189 and 1190 as instructed below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1189 PP1-J78 5113186-2	RUN #1329 PP1-J78 5160765	RUN #1189 MR2-DVR-J5 5113186-2	RUN #1329 MR2-DVR-J10 5160765
RUN #1190 PP1-J203 5113186-5	RUN #1330 PP1-J203 5160765-4	RUN #1190 MR2-DVR-J10 5113186-5	RUN #1330 MR2-DVR-J5 5160765-4

4-4 CABLE MAPS

The Affected (removal) cable maps are located on the following pages. Select the applicable map for site being upgraded.

- Affected (Removed) Cables for Upgrade to HDx/HDxt from Upgraded Excite HD 1.5T Page 19
- Affected (Removed) Cables for Upgrade to HDx/HDxt from Excite HD 1.5T Page 20
- Affected (Removed) Cables for Upgrade to HDx/HDxt from Upgraded Excite HD 3.0T Page 21
- Affected (Removed) Cables for Upgrade to HDx/HDxt from Excite HD 3.0T Page 22

The Affected (removal) cable maps are located on the following pages. Select the applicable map for site being upgraded.

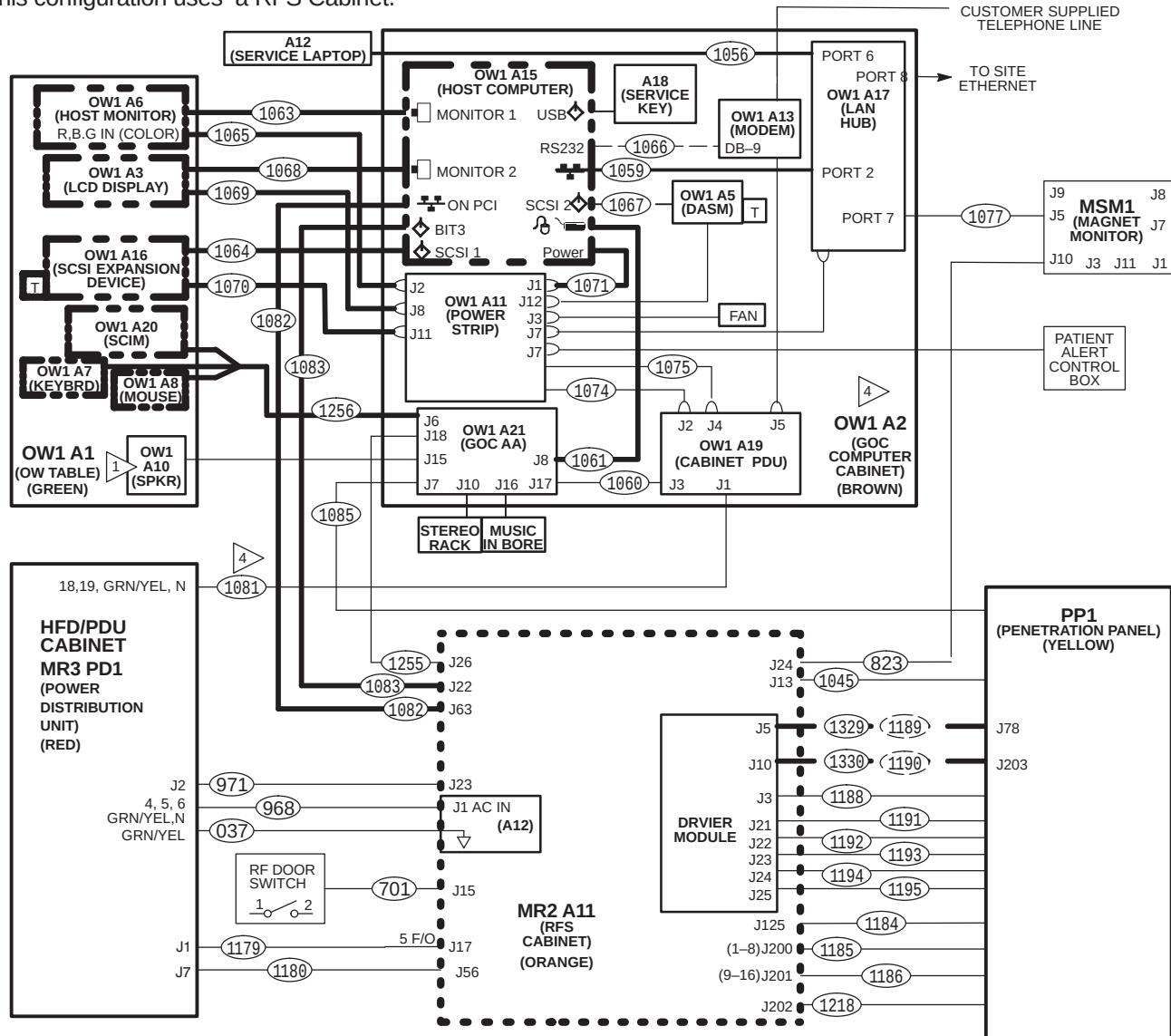
- Affected (Added) Cables for Upgrade to HDx/HDxt from Upgraded Excite HD 1.5 Page 23
- Affected (Added) Cables for Upgrade to HDx/HDxt from Excite HD 1.5T Page 24
- Affected (Added) Cables for Upgrade to HDx/HDxt from Upgraded Excite HD 3.0T Page 25
- Affected (Added) Cables for Upgrade to HDx/HDxt from Excite HD 3.0T Page 26

The cable map represents a site that was upgraded to EXCITE HD prior to the upgrade being performed now. This configuration uses a System Cabinet.



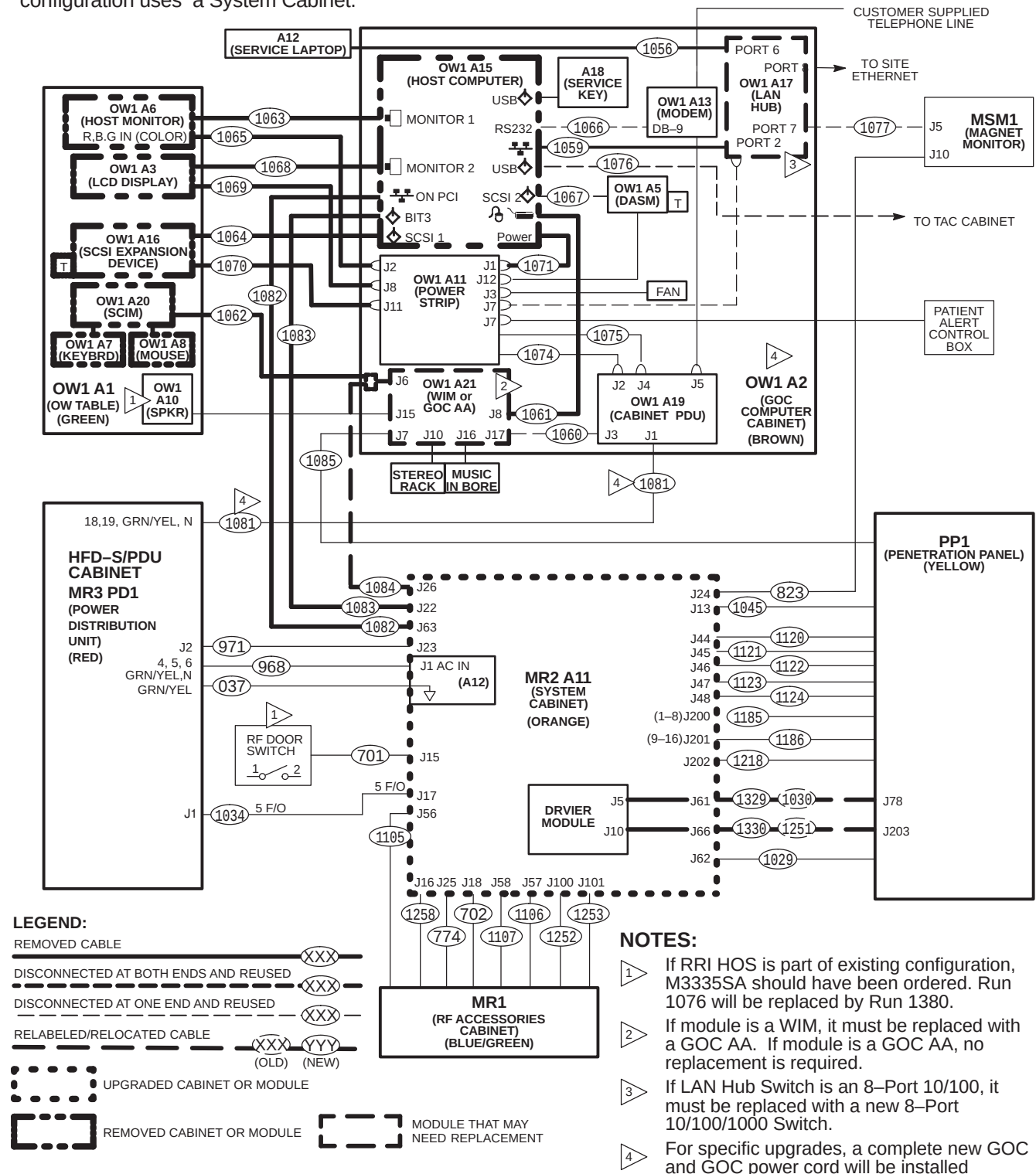
4-4-2 AFFECTED (REMOVED) CABLES FOR UPGRADE TO HDx/HDxt FROM EXCITE HD 1.5T

The cable map represents a site that was installed new as EXCITE HD prior to the upgrade being performed now. This configuration uses a RFS Cabinet.



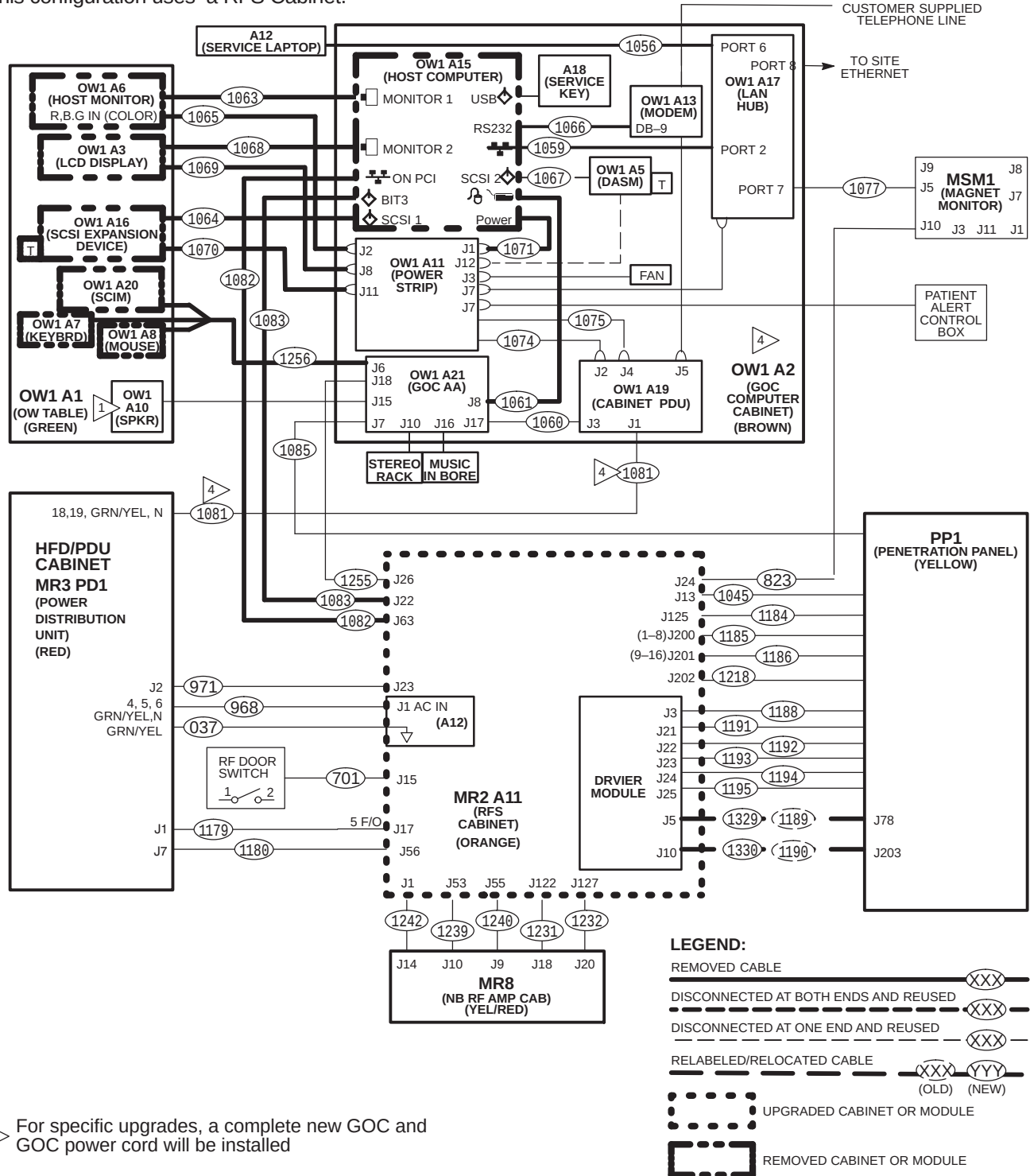
4-4-3 AFFECTED (REMOVED) CABLES FOR UPGRADE TO HDxHDxt FROM UPGRADED HD 3.0T

The cable map represents a site that was upgraded to EXCITE HD prior to the upgrade being performed now. This configuration uses a System Cabinet.



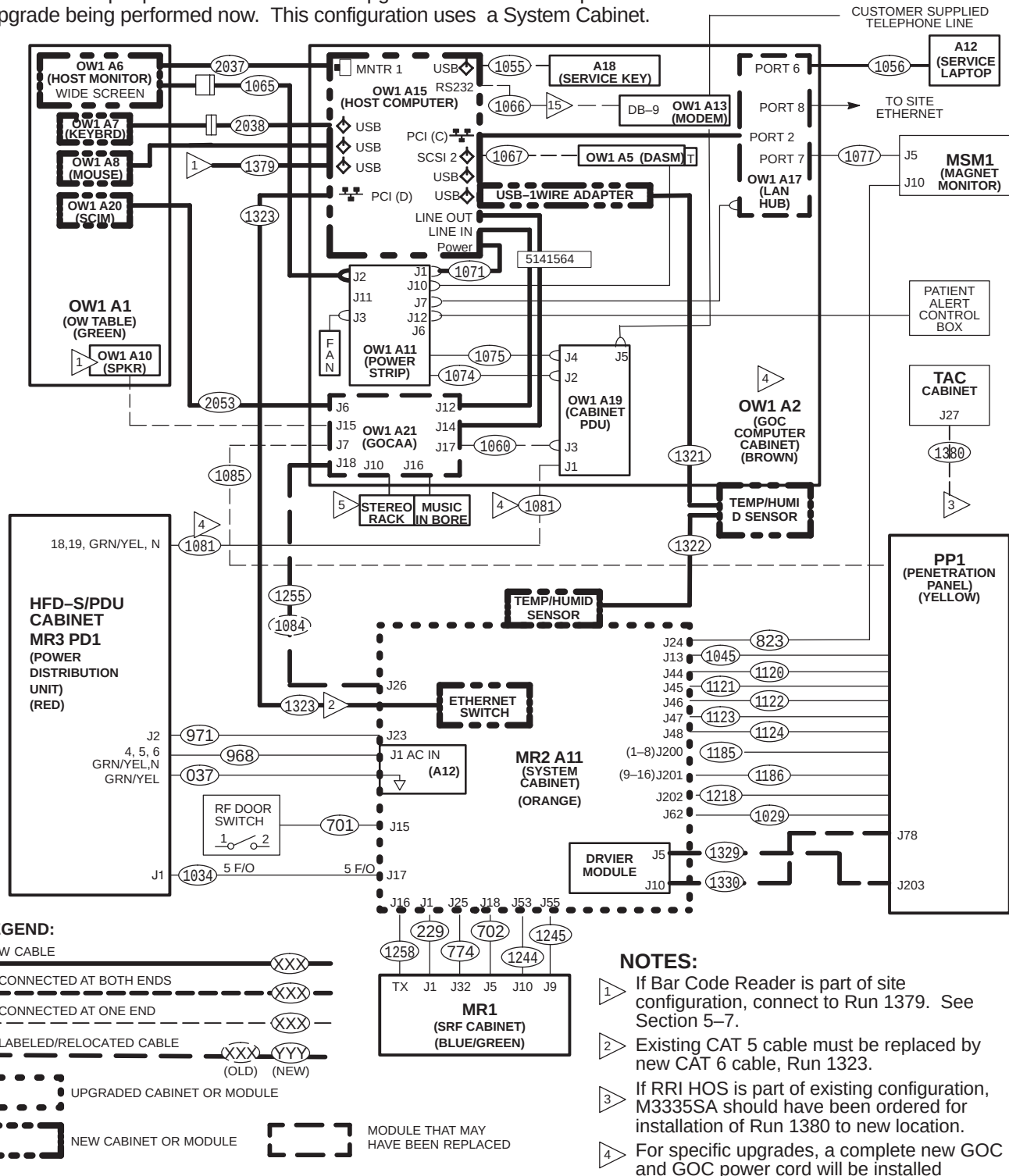
4-4-4 AFFECTED (REMOVED) CABLES FOR UPGRADE TO HDx/HDxt FROM EXCITE HD 3.0T

The cable map represents a site that was installed new as EXCITE HD prior to the upgrade being performed now. This configuration uses a RFS Cabinet.



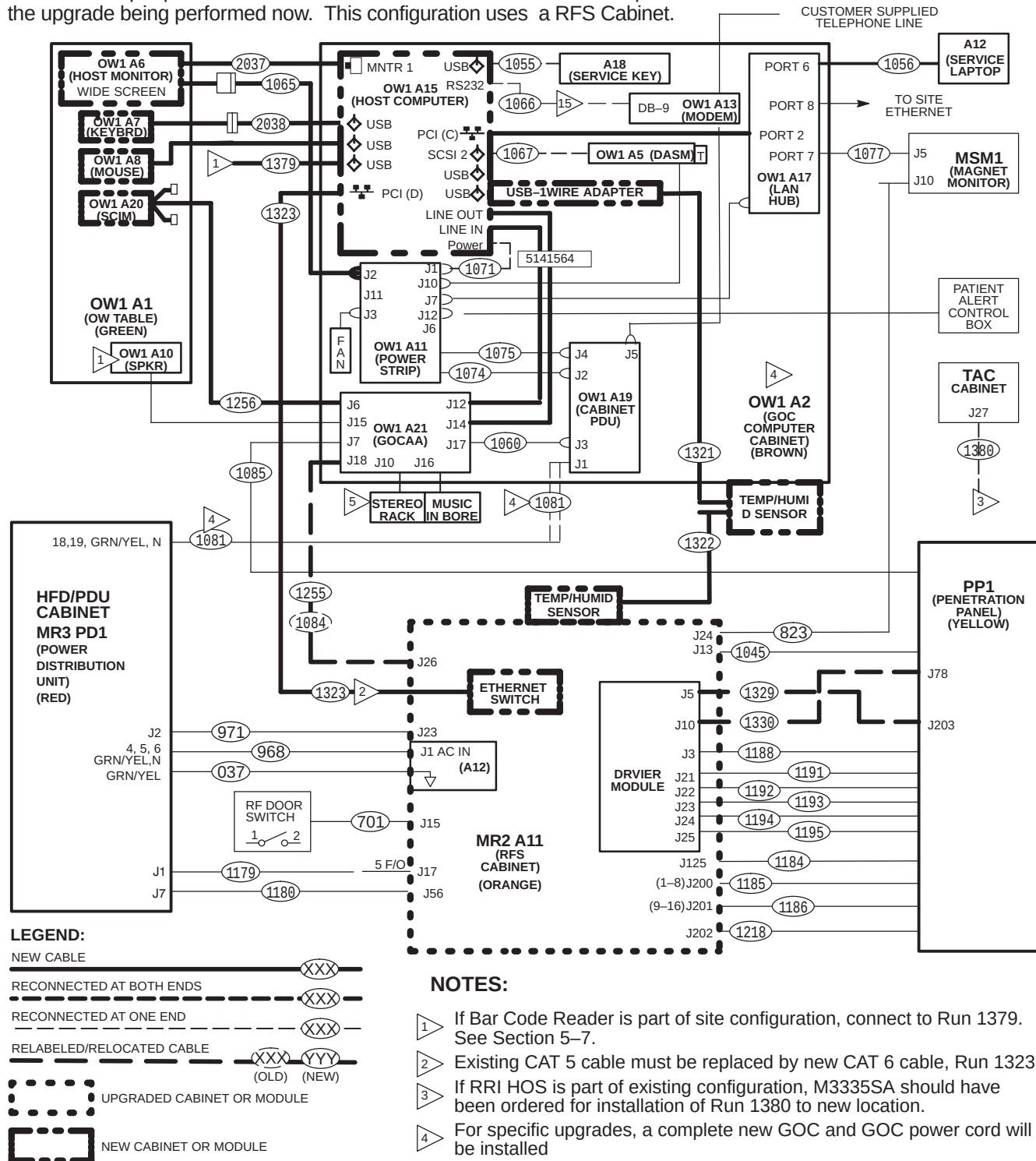
4-4-5 AFFECTED (ADDED) CABLES FOR UPGRADE TO HDx/HDxt FROM UPGRADED EXCITE HD 1.5T

The cable map represents a site that was upgraded to EXCITE HD prior to the upgrade being performed now. This configuration uses a System Cabinet.



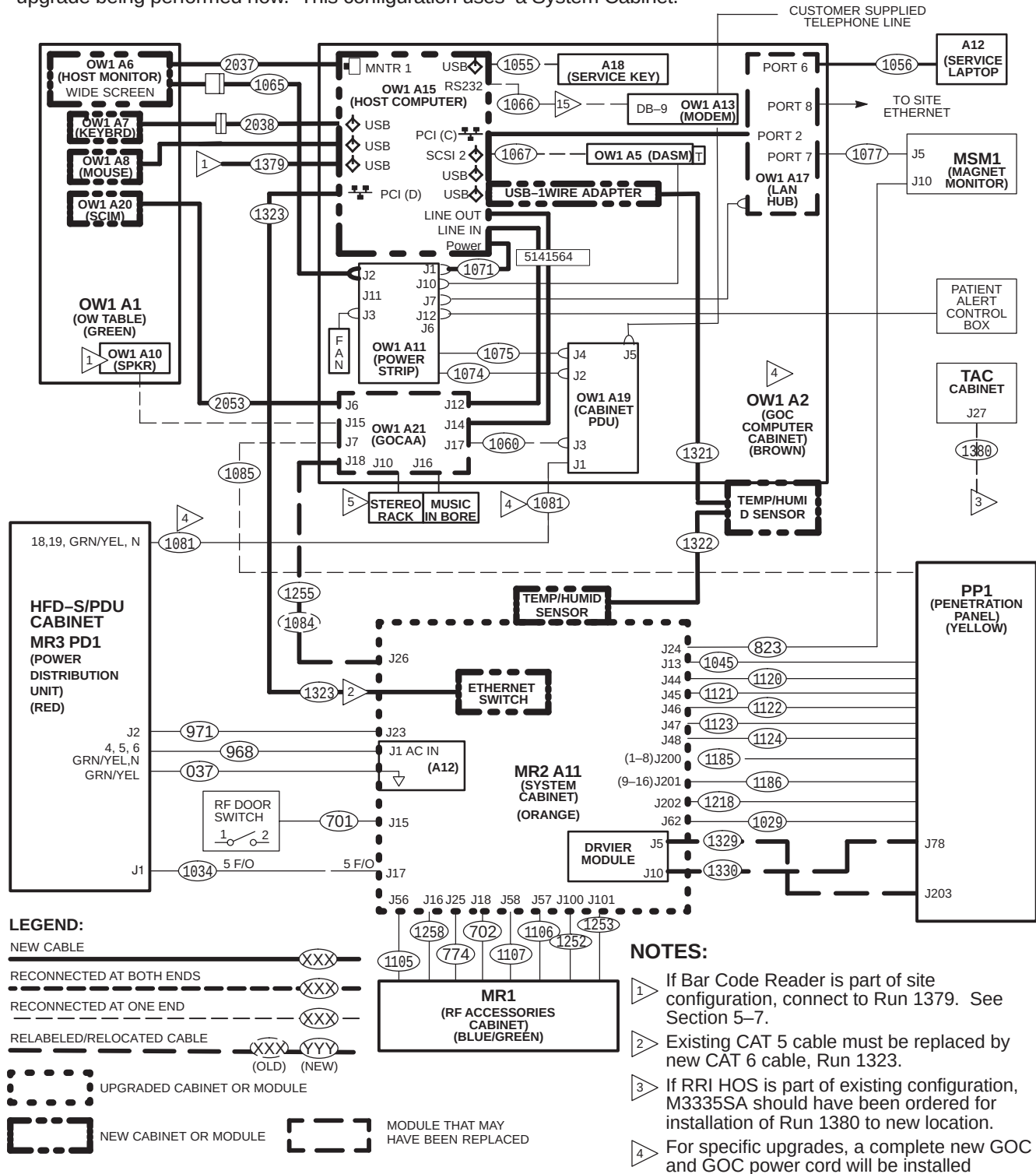
4-4-6 AFFECTED (ADDED) CABLES FOR UPGRADE TO HDx/HDxt FROM EXCITE HD 1.5T

The cable map represents a site that was installed new as EXCITE HD prior to the upgrade being performed now. This configuration uses a RFS Cabinet.



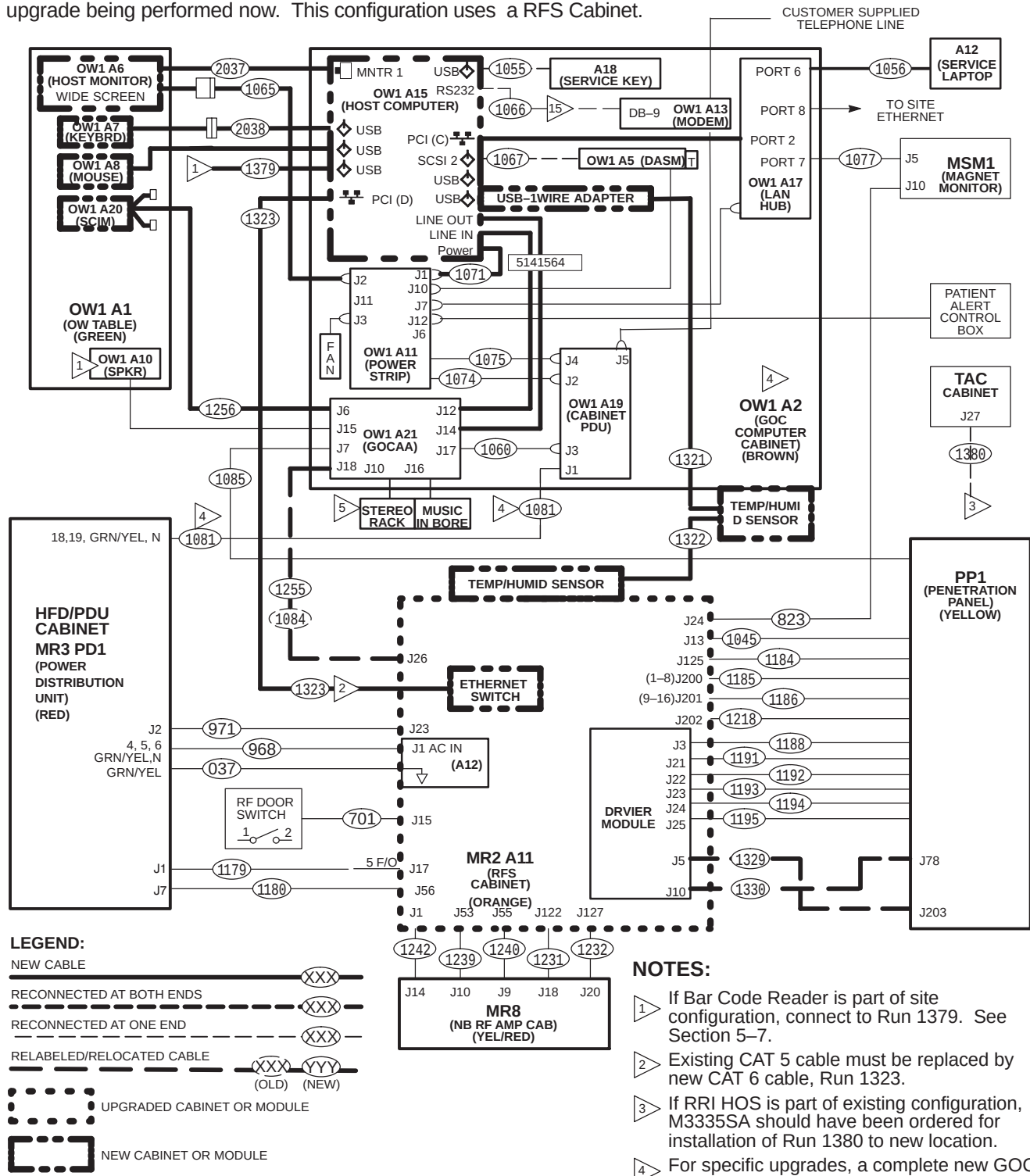
4-4-7 AFFECTED (ADDED) CABLES FOR UPGRADE TO HDx/HDxt FROM UPGRADED EXCITE HD 3.0T

The cable map represents a site that was upgraded to EXCITE HD prior to the upgrade being performed now. This configuration uses a System Cabinet.



4-4-8 AFFECTED (ADDED) CABLES FOR UPGRADE TO HDx/HDxt FROM EXCITE HD 3.0T

The cable map represents a site that was installed new as EXCITE HD prior to the upgrade being performed now. This configuration uses a RFS Cabinet.



SECTION 5 – OPERATOR WORKSPACE UPGRADE

5–1 INTRODUCTION

A new Monitor, SCIM, and Keyboard will replace existing components on the Operator Workspace Table. Also, the new Temperature and Humidity kit will be installed along with the new Z400 computer.

A new Z400 computer is also installed for this upgrade. Depending on site requirements, there are two options for this computer upgrade. They are:

- A replacement of the existing GOC with a new GOC containing the Z400 computer. Refer to Section 5–2.
- A replacement of only the existing computer with the new Z400 computer. Refer to Section 5–3.

After completing one of the above options, continue the Operator Workspace upgrade with Section 5–4.

5-2 INSTALL NEW GOC CONTAINING Z400 COMPUTER

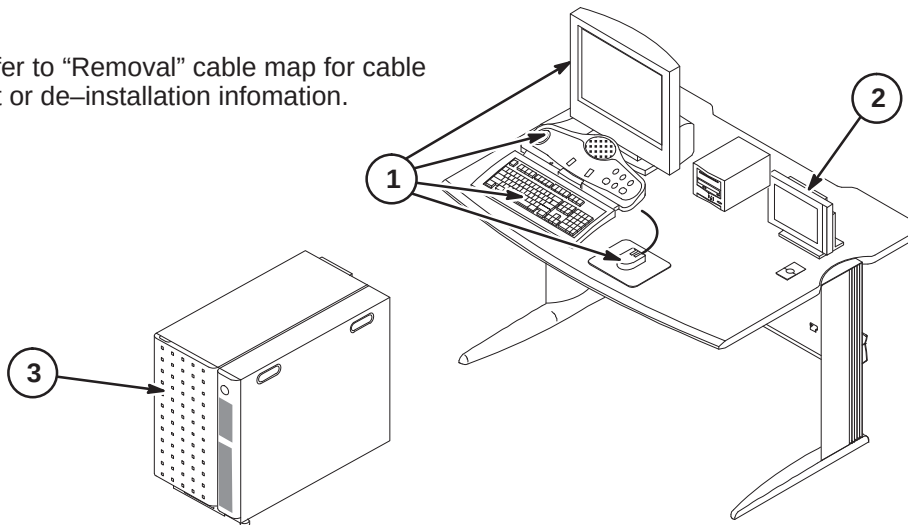
5-2-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

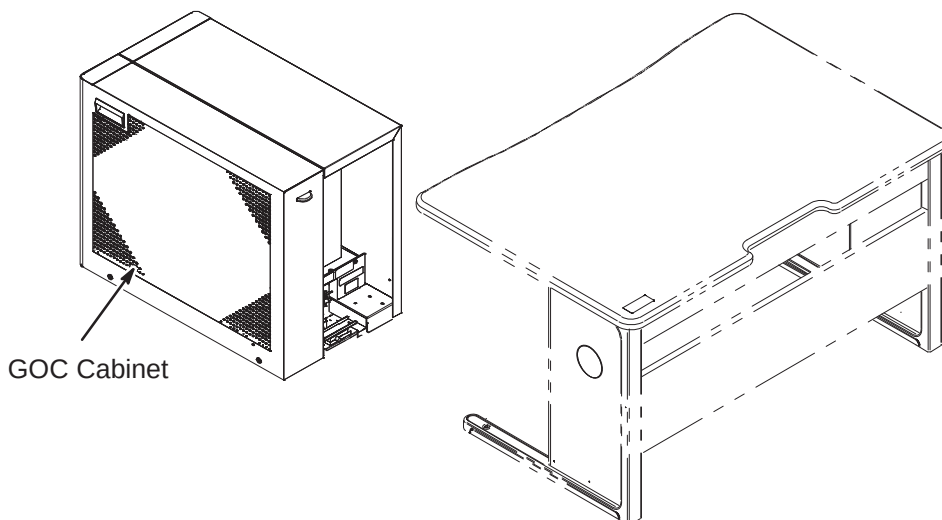
1. Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.
Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.
2. De-install LCD Display, Runs 1068 and 1069, and return for recycling.
3. Existing GOC will be replaced by a new GOC containing the Z400 computer:

Note: Refer to "Removal" cable map for cable retainment or de-installation information.



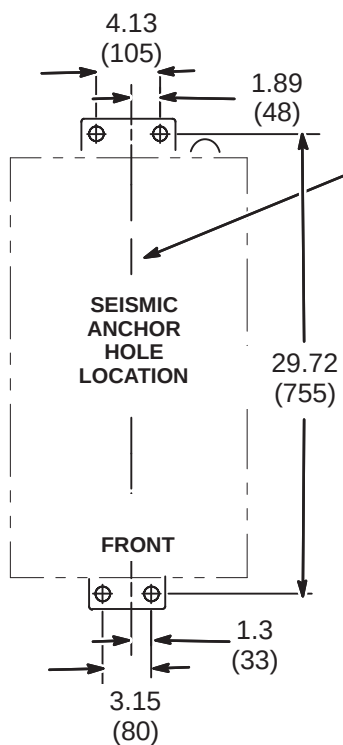
5-2-2 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



5-2-3 SEISMIC ANCHORING (IF REQUIRED)

GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

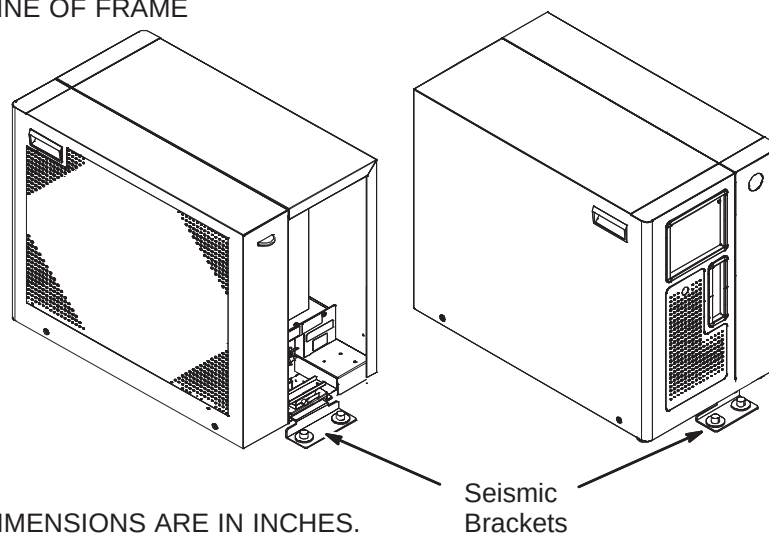


If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets

CENTERLINE OF FRAME

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



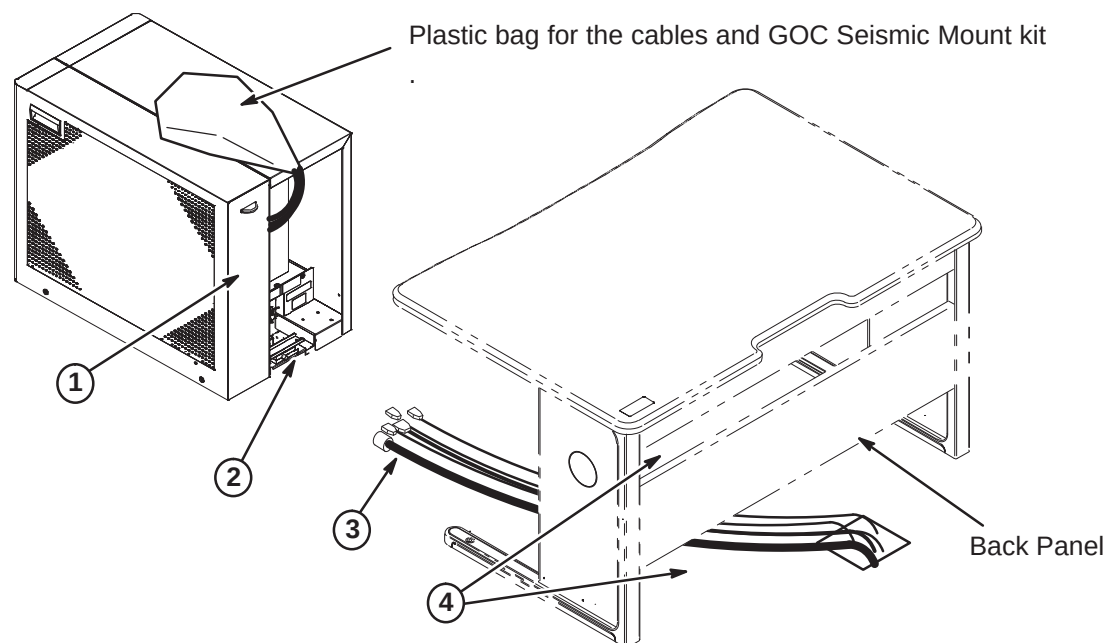
5-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video output cable
- Keyboard output cable
- SCIM output cable
- SCSI output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening.
(These cables are inside of plastic bag)
2. Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



5-2-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

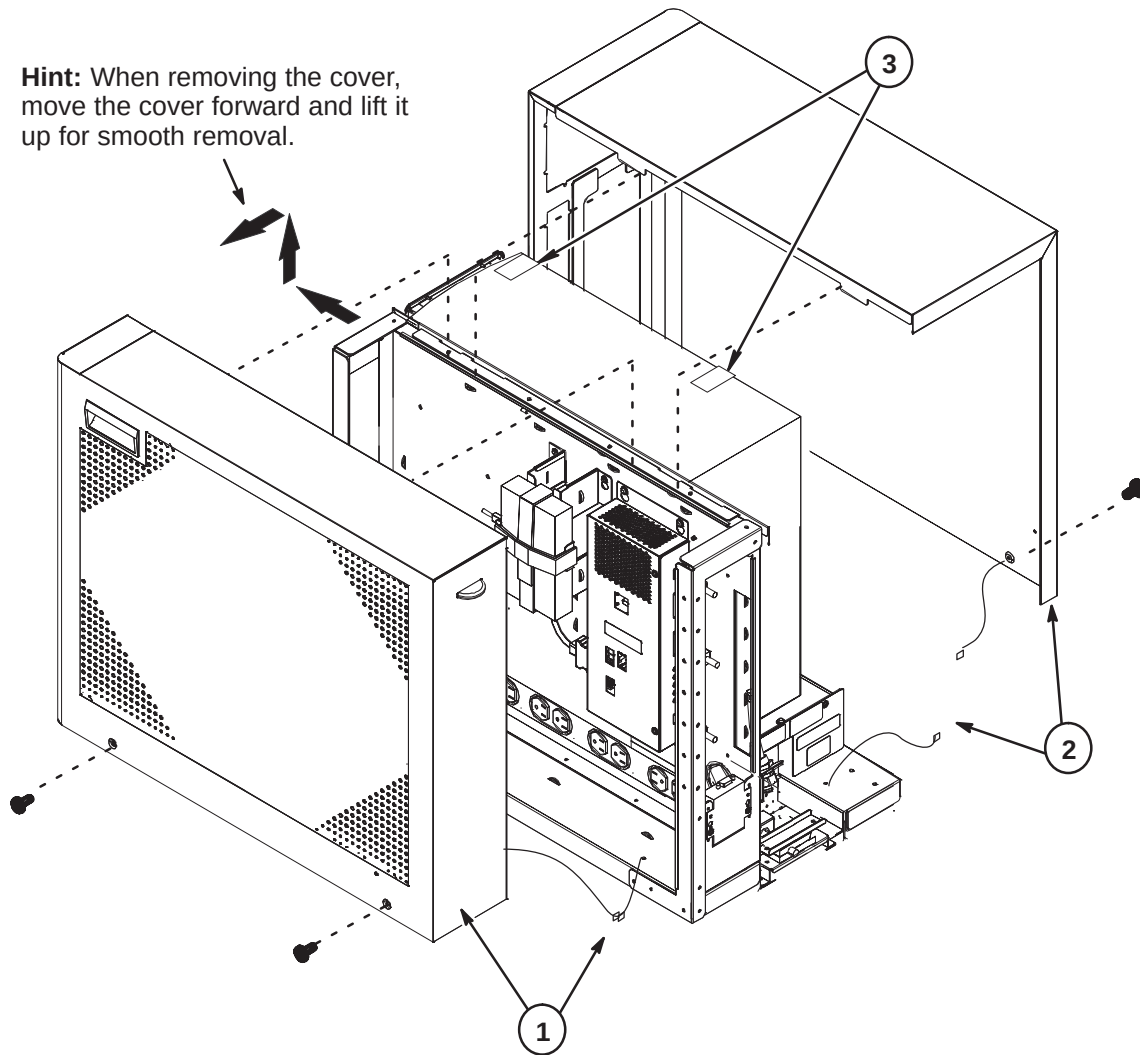
1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

3. Peel the two shipping tapes off from PC.

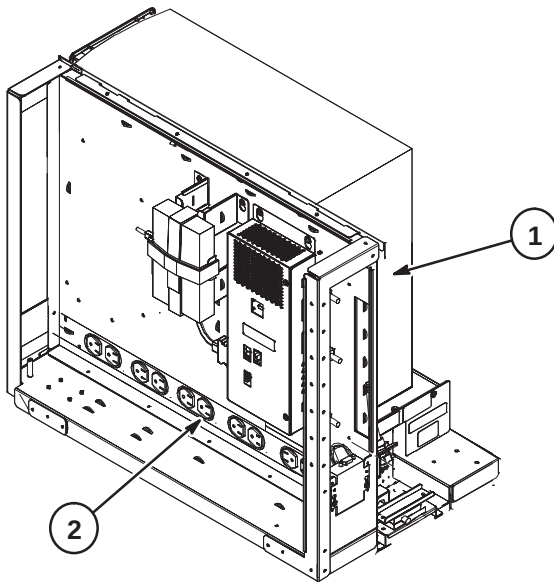
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



5-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



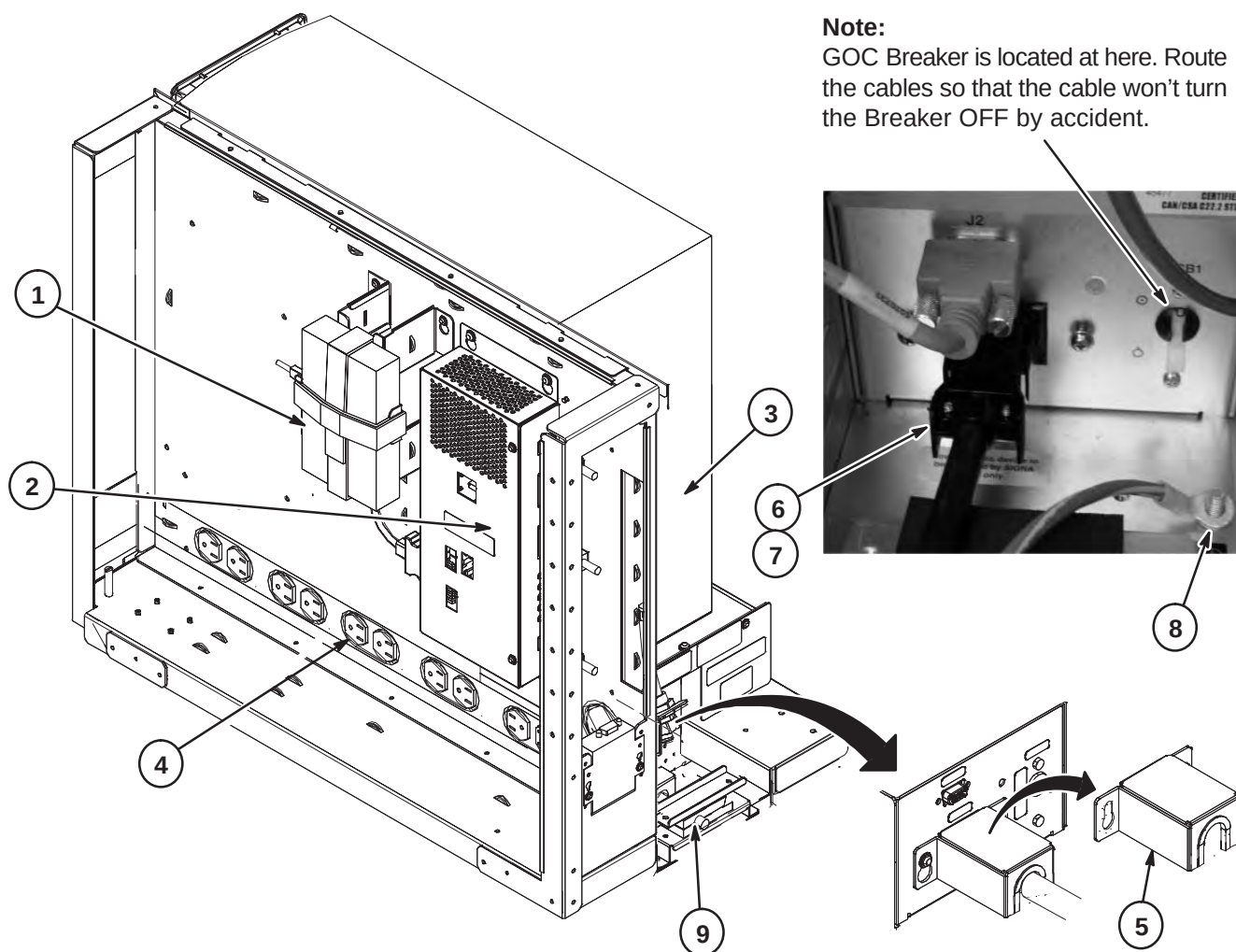
5-2-7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

1. Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

2. Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
3. Refer to Section 8-2-9 for all cables connected to Linux PC computer.
4. Refer to Section 5-2-8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
5. Before connecting GOC power cable Run 1381 to J1, loosen screws and remove cover.
6. Route and connect GOC power cable Run 1381 to J1 on PDU.
7. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5. after connecting cable and tighten screws.
8. Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
9. Tighten clamp to secure cables in place.

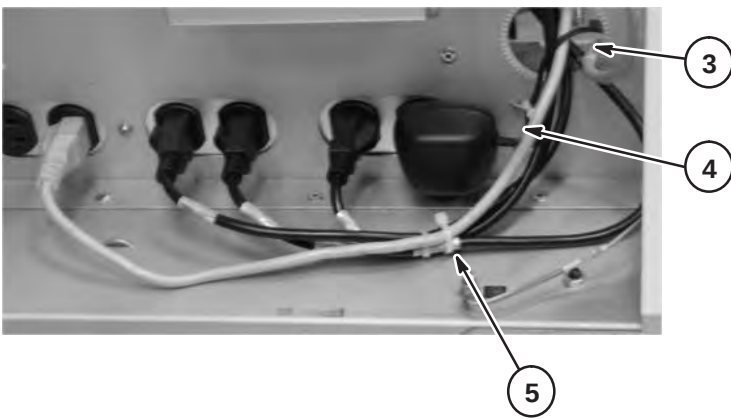
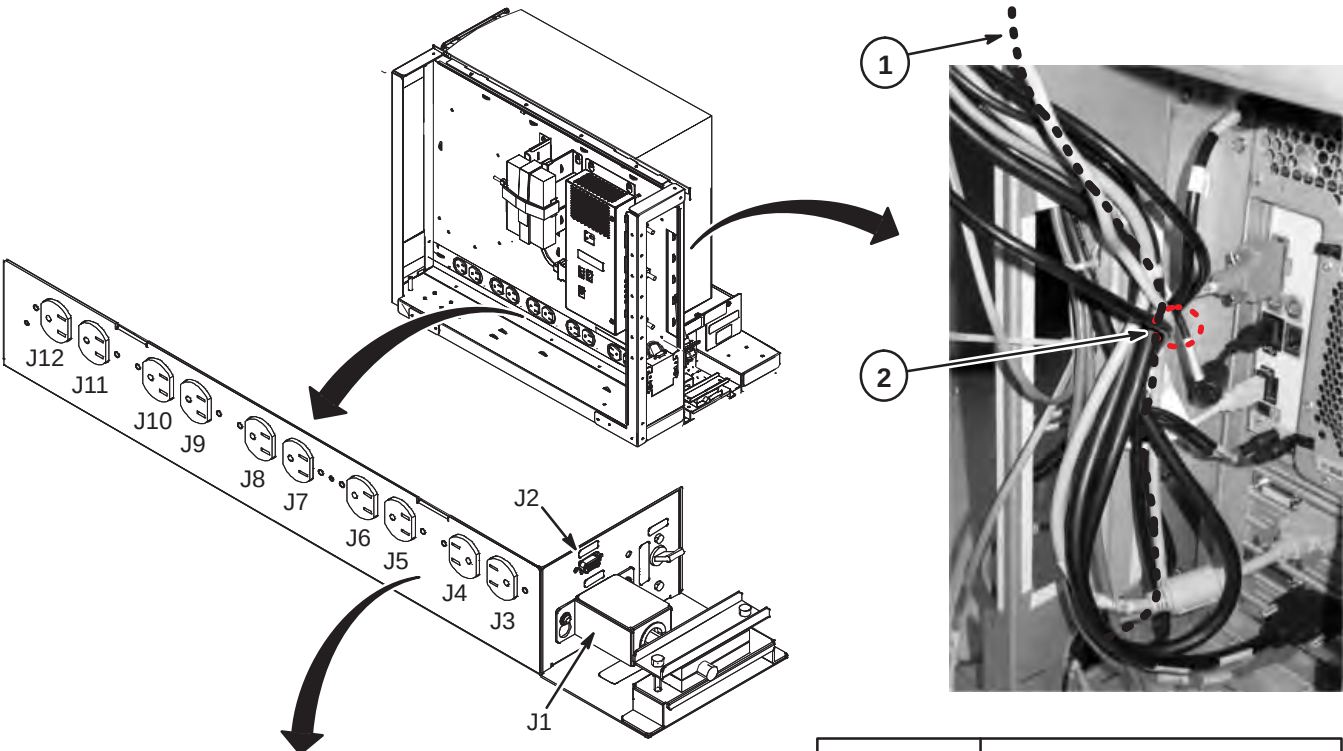


5-2-8 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

- 1. If Modem is installed, route power cable and connect to J7.
- 2. Tie-wrap point for Modem power cable.
- 3. Route power cables connected to J3 thru J7 through access hole.
- 4. route and connect Patient Alert power cable to J3 and tie-wrap at this point.
- 5. Tie-wrap power cables J4 to J7 at this point.



POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Computer
J5	Host LCD
J6	SCSI Tower
J7	Modem
J8	
J9	LAN Hub
J10	
J11	
J12	One Wire Hub

5-3 INSTALL NEW Z400 COMPUTER IN EXISTING GOC

5-3-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

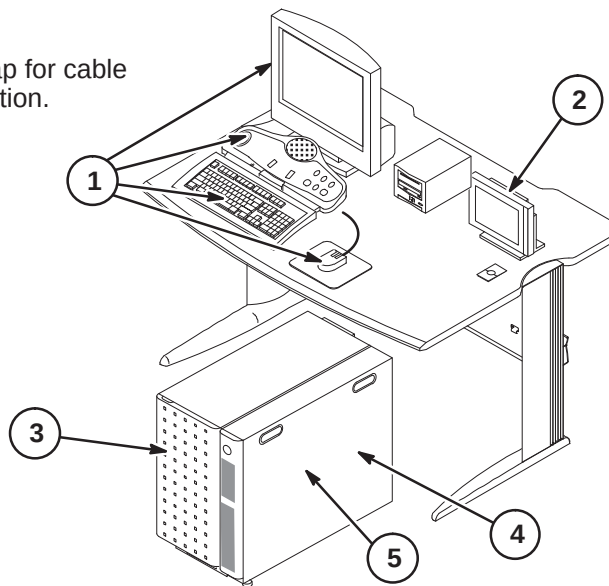
Make sure that all system applications are shut down and that power at the PDU is turned off and LOTO applied.

1. Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.

Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.

2. De-install LCD Display, Runs 1068 and 1069, and return for recycling.
3. Existing PC computer in the GOC will be replaced by a new computer:
4. The GOC AA module has been delivered as part of the upgrade kit.
 - If the existing GOC already has a GOC AA module installed, the new GOC AA is not required.
 - If the existing GOC has a WIM installed, it must be replaced by the new GOC AA module.
5. An 8-Port 10/100/1000 Mbps Gigabit Ethernet Switch has been delivered as part of the upgrade kit.
 - If the existing GOC already has this Switch installed, the new Switch is not required.
 - If the existing LAN Hub Switch is an 8-Port 10/100, then it must be replaced by the new 10/100/1000 Switch.

Note: Refer to "Removal" cable map for cable retainment or de-installation information.

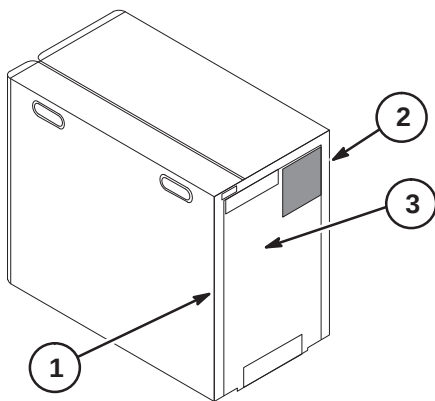


5-3-2 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

5-3-2-1 OLDER STYLE GOC COVERS

1. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
2. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
3. Loosen four screws holding the back cover. Lift and remove cover.

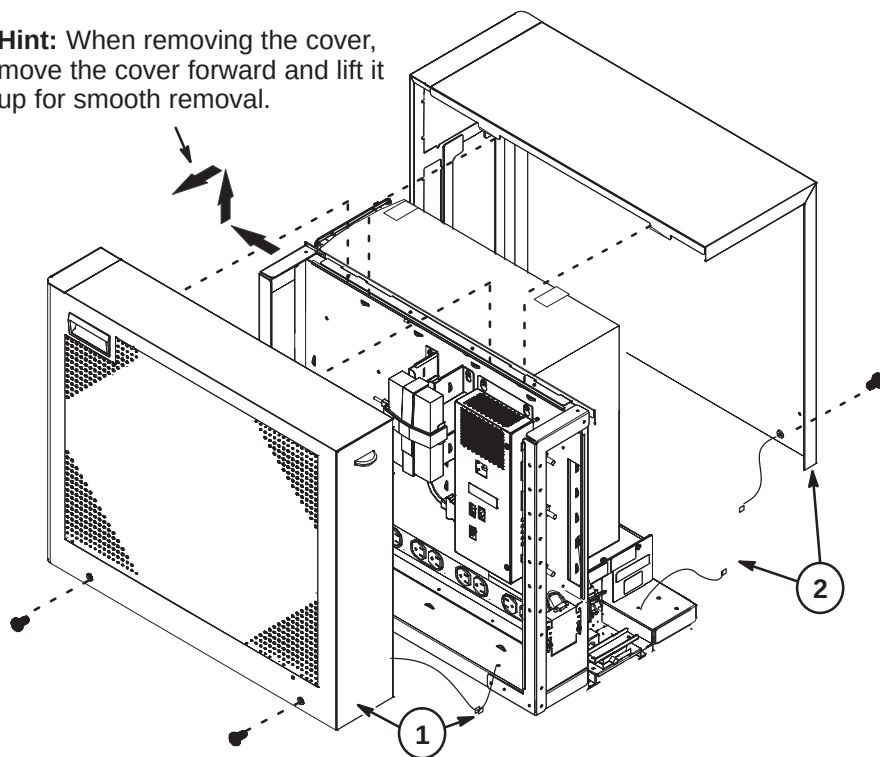


5-3-2-2 NEWER STYLE GOC COVERS

Note: A ground wire is connected to each side cover. Use care when removing side covers.

1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

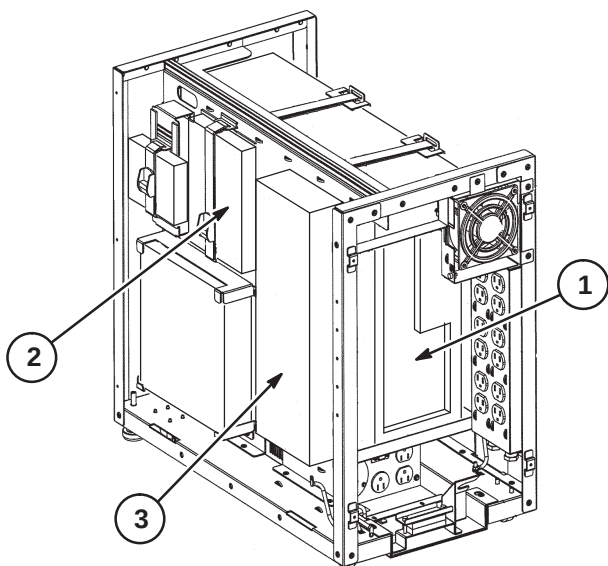
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



5-3-3 GOC COMPONENT REPLACEMENT

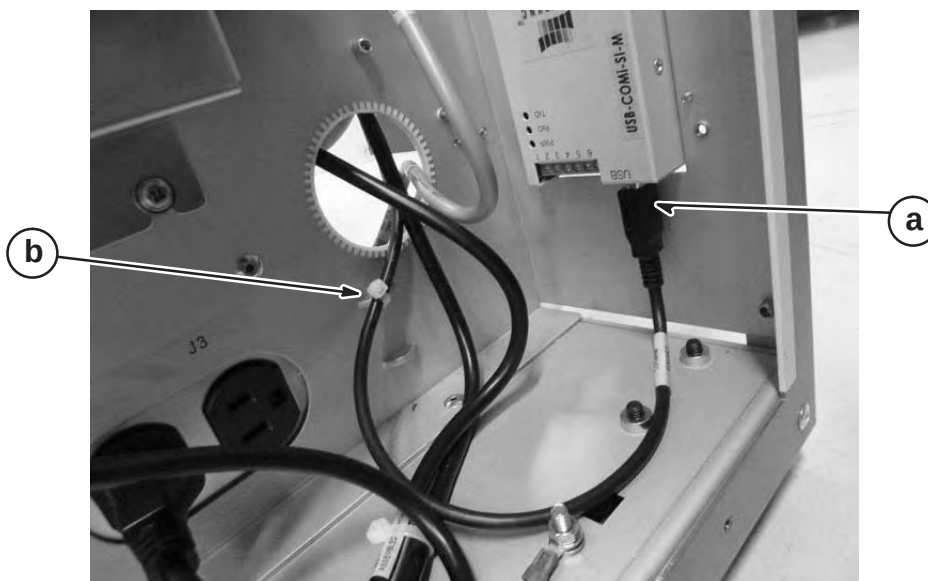
Refer to applicable cable removal map for de-installation of existing cables that will be replaced by new cables. The cable maps also show which cables will be disconnected from existing components and reconnected to new components.

1. Disconnect all cables from existing computer. Make sure they are properly labeled.
2. If required, replace existing LAN Hub Switch with new 10/100/1000 Mbps Gigabit Ethernet Switch (5139480-2).
3. If required, replace existing WIM with new GOC AA module (2395043).



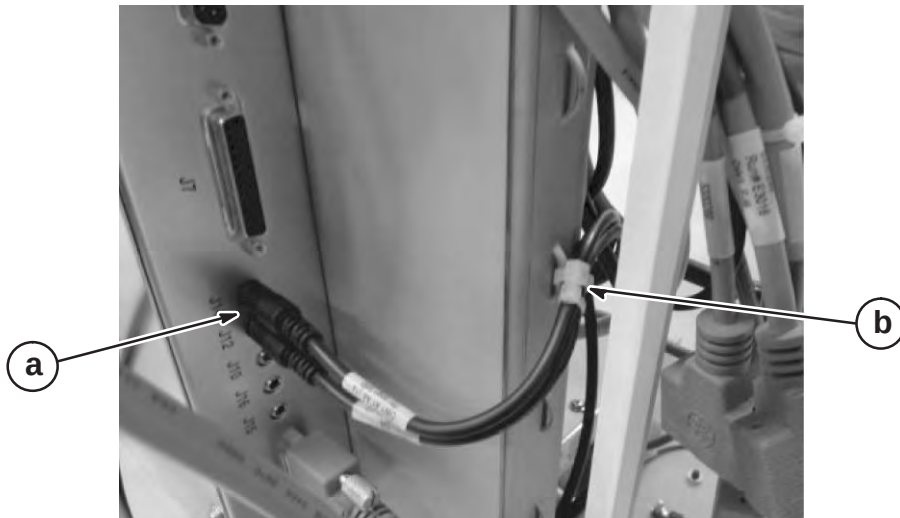
5-3-4 INSTALLATION OF NEW GOC COMPONENTS

1. Replace the Host to USB-Serial Converter USB Cable.
 - a. Reconnect the Host to USB-Serial Converter USB Cable.
 - b. Cut and restore tie-wrap(s) as required during cable replacement.

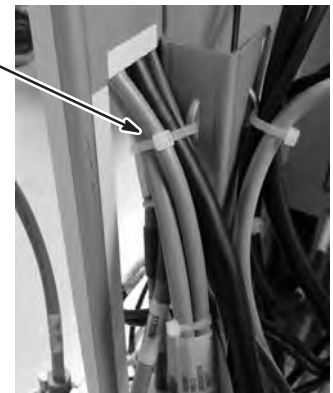
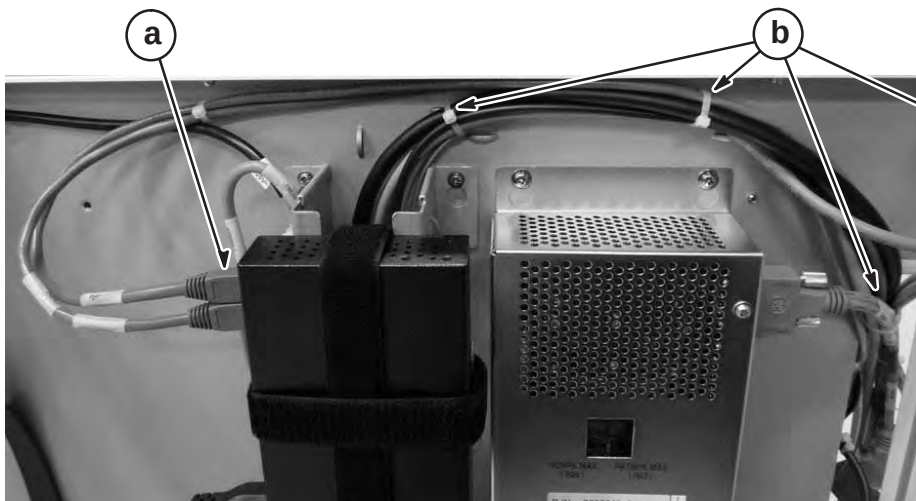


5-3-4 INSTALLATION OF NEW GOC COMPONENTS (Continued)**2. Replace the Audio Cable Assembly.**

- a. Reconnect the Audio Cable Assembly, connected to J12 and J14 on GOCAA.
- b. Cut and restore tie-wrap(s) as required during cable replacement.

**3. Replace the Host-Hub Ethernet Cable.**

- a. Reconnect the Ethernet Cable between the Host and Port 1 of Ethernet Switch Hub.
- b. Cut and restore tie-wrap(s) as required during cable replacement.



View from GOC rear side.

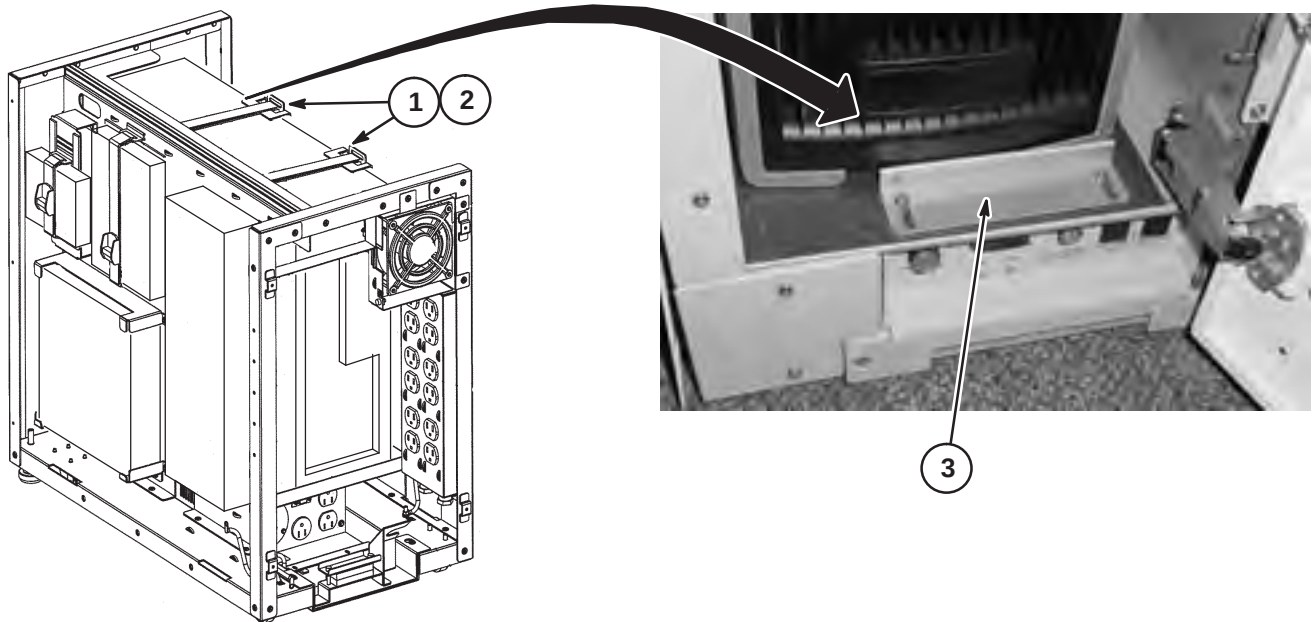
5-3-5 GOC COMPUTER REPLACEMENT**IMPORTANT!**

XW8000, 8200, 8400, or 9300 PC internal hardware, such as any hard drive, is not compatible with the Z400 computer. Do not attempt to transfer any internal hardware between any of these and the Z400 computer.

CAUTION

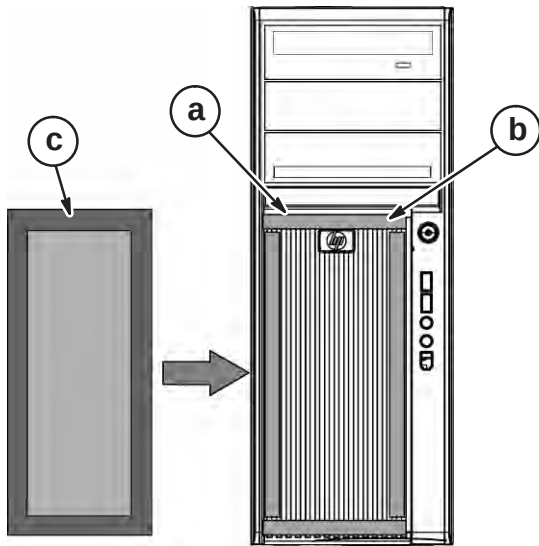
The computer weighs about 50 pounds (23Kg). This requires two people to handle the computer.

1. Unfasten straps and carefully remove computer.
2. Because the new Z400 computer is shorter than the original PC being replaced, install the new computer to the rear-most position and secure in place with straps.
3. If required, install PC Stopper Bracket (5127954) at front of computer using existing screws.



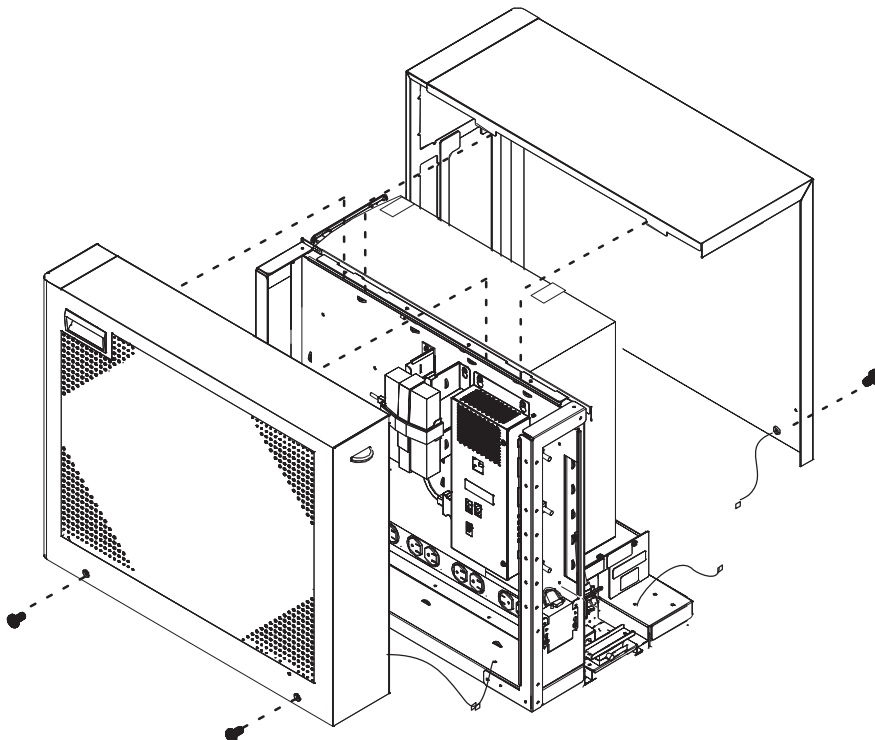
5-3-5 GOC COMPUTER REPLACEMENT (Continued)

4. If applicable, attach the Dust Filter to the front panel after following the cleaning instructions below.
 - a. Clean the velcro area of front panel with alcohol and cloth. Peel the velcro from backside of filter.
 - b. Attach the velcro around front panel.
 - c. Attach the filter to front panel.



IMPORTANT! Make sure the Z400 computer is installed at the rear-most position before installing covers. This position will aid in connecting existing cables to rear of Z400 computer.

5. Reinstall the GOC covers.



5-4 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION

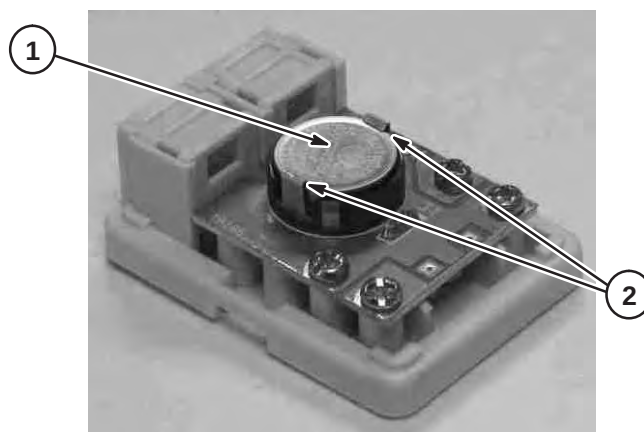
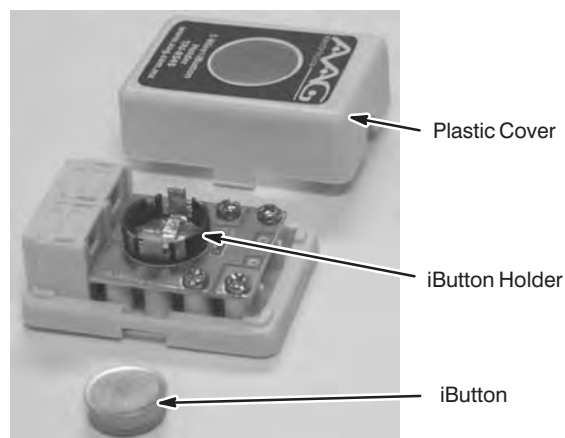
The components of the Temperature and Humidity Sensor Kit (5147136-2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

5-4-1 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

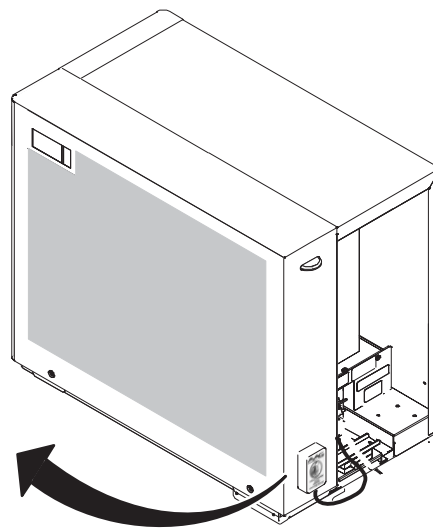
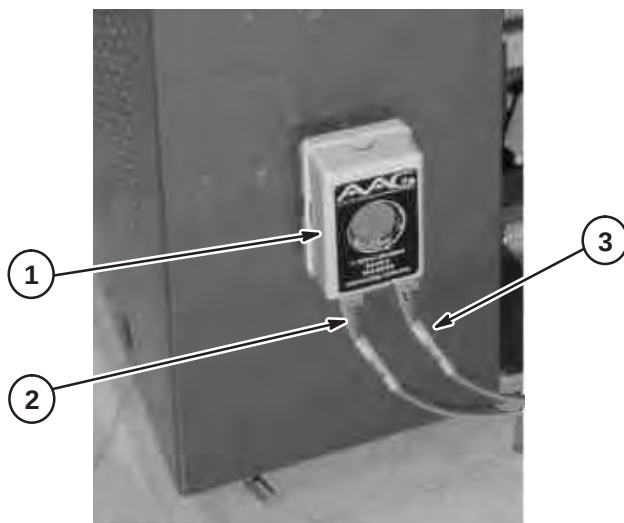
Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

1. Install the iButton in the holder so that the solid part of the case is exposed.
2. Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
3. If removed, reinstall the Plastic Cover.



5-4-2 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC

1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857-5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Route and connect Run 1322 (5143794-2) from System cabinet to Sensor at back of GOC.
3. Connect cable Run 1321 (5143794-2) to Sensor and route other end thru bottom opening of GOC and connect to USB to 1-Wire Adapter (5307784) installed in GOC USB port. Refer to Section 5-3-4.



5-4-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

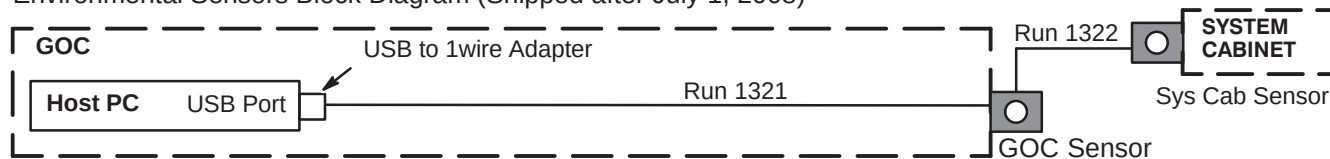
Note: iButton Holders are unique to each location. Make sure to install 5141857-4 to System Cabinet and 5141857-5 to GOC.

1. Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857-4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Route and connect cable Run 1322 (5143794) from GOC rear cover.

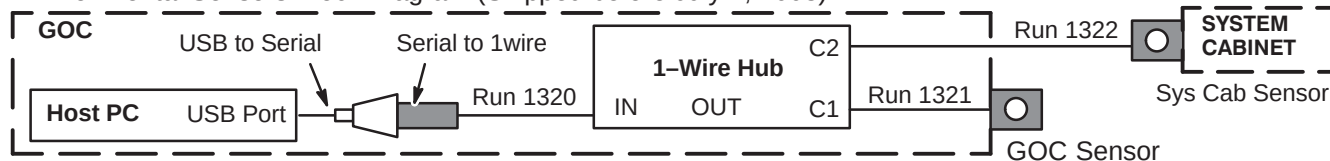


5-4-4 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after July 1, 2008)



Environmental Sensors Block Diagram (Shipped before July 1, 2008)

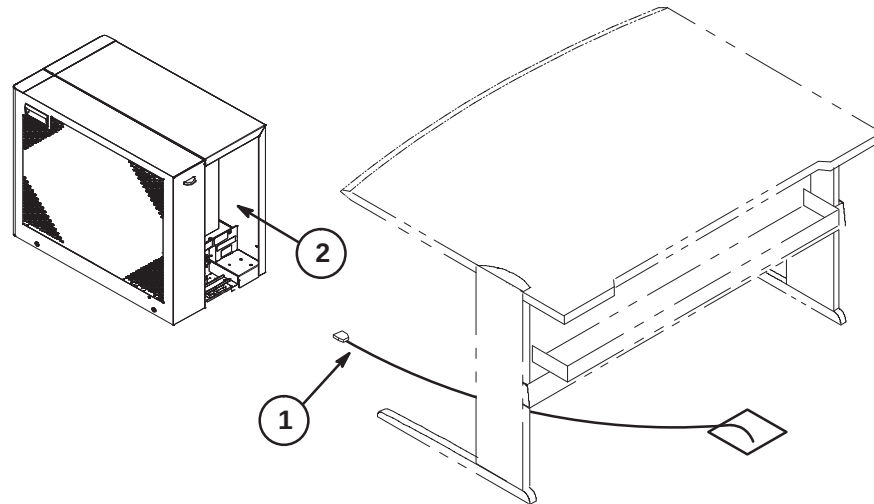


5-5 ROUTE RUN 1323

1. Route Run 1323 from RFS or System Cabinet to location of GOC. This is a new CAT6 cable that must be installed. Existing CAT5 cables cannot be reused.

NOTE: For optimal efficiency, route Run 1322, Section 5-4-3, at the same time as Rune 1323. It is also routed between RFS/System Cabinet and GOC.

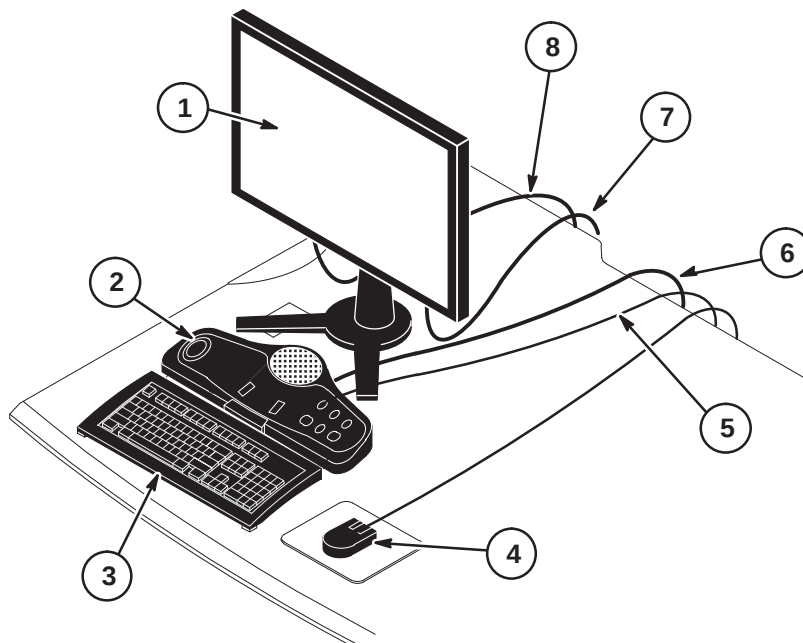
2. Cable will be routed thru the opening to the PC for connection.



5-6 INSTALL NEW OW TABLE TOP COMPONENTS

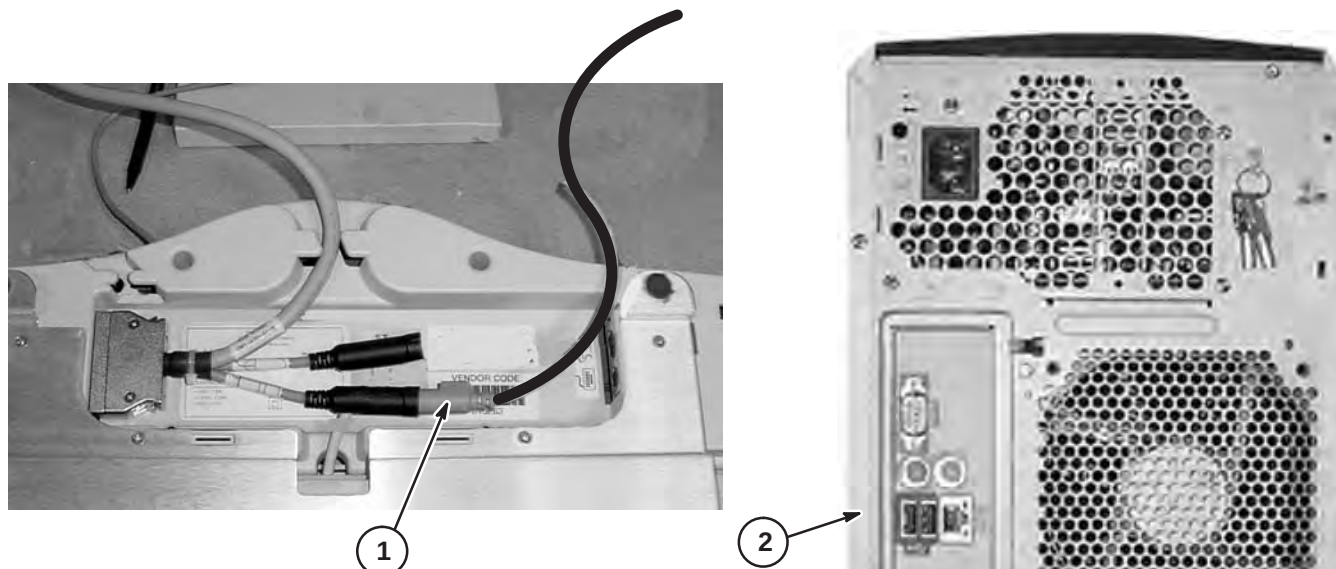
1. Install new Monitor on table.
2. Place new SCIM (5148388-x) on table.
3. Place new Keyboard (5143798-x) on table.
4. Install new Mouse (5143798-6) on table and route USB cable to computer in GOC. Attach provided extension USB cable if necessary.
5. Connect extension cable Run 2038 to USB Keyboard cable and route to computer in GOC.
6. Connect Run 2053 (or 1256) to SCIM and route to rear of GOC.
7. Connect provided monitor power cable to monitor. Connect other end of cable to Run 1065, which is connected to OW power strip.
8. Connect DVI cable, Run 2037, to monitor and route to rear of computer in GOC.

Note: Do not use or connect provided VGA cable.

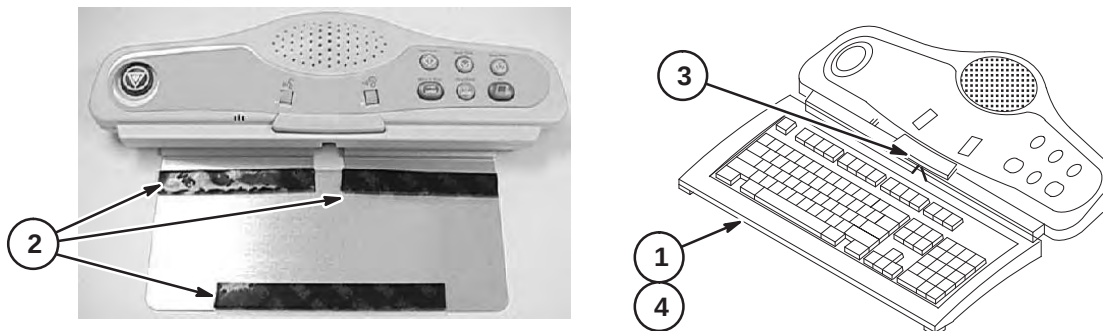


5-7 CONNECT BAR CODE READER (If part of site configuration)

1. *If Run1256 is present* – Disconnect the Bar Code Reader cable from Run 1256 Y-connector under SCIM.
2. Connect cable Run 1379 to rear of computer and Bar Code Reader.

**5-8 INSTALL SCIM/KEYBOARD**

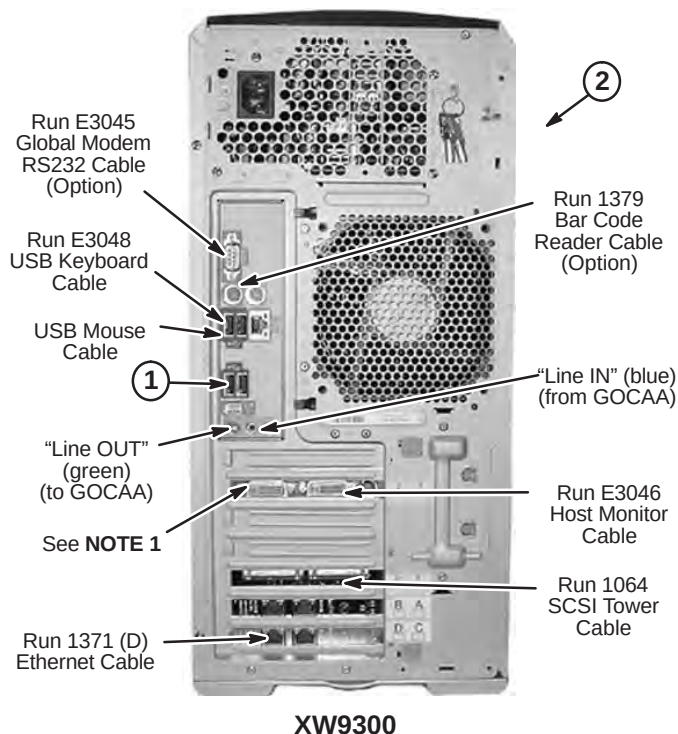
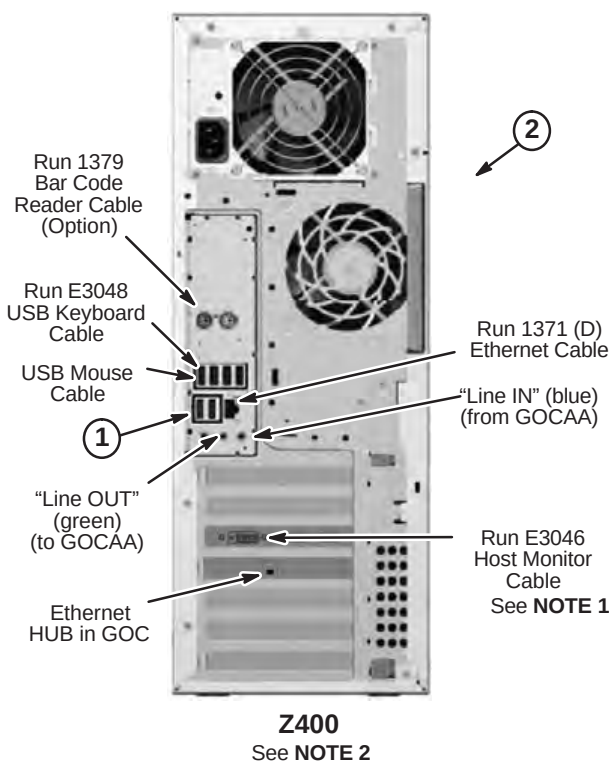
1. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
2. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface.
3. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
4. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

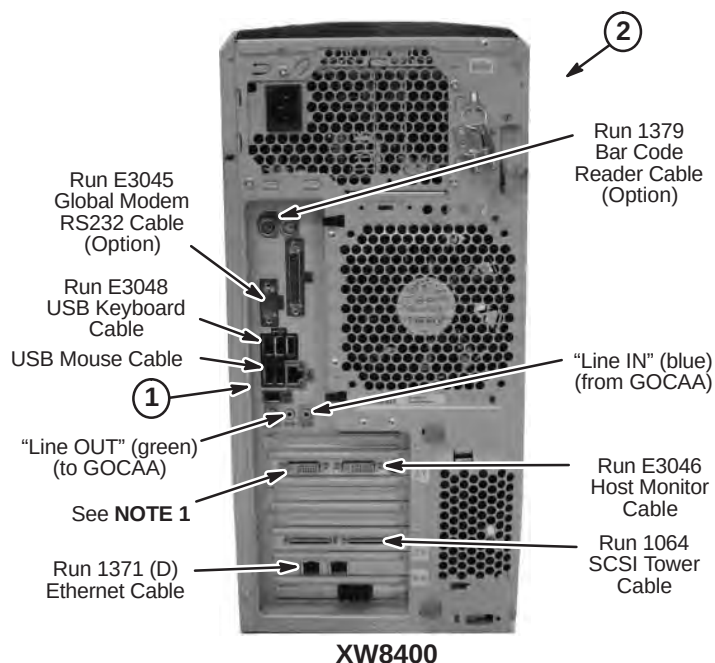
5-9 CONNECT CABLES TO COMPUTER

1. Refer to Section 5-4 for installing the USB to 1-Wire Adapter (5307784) to USB port.
2. Connect remaining cables as marked at rear of computer.



NOTE 1:
Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot-up to stop.

NOTE 2:
SCSI Tower and DASM are not supported for Z400PC.



5-10 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

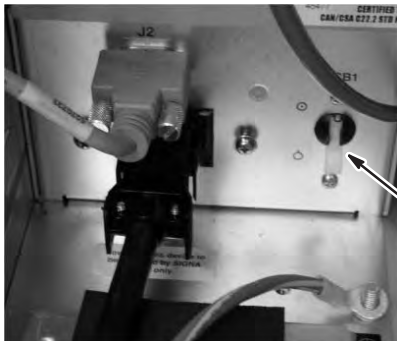
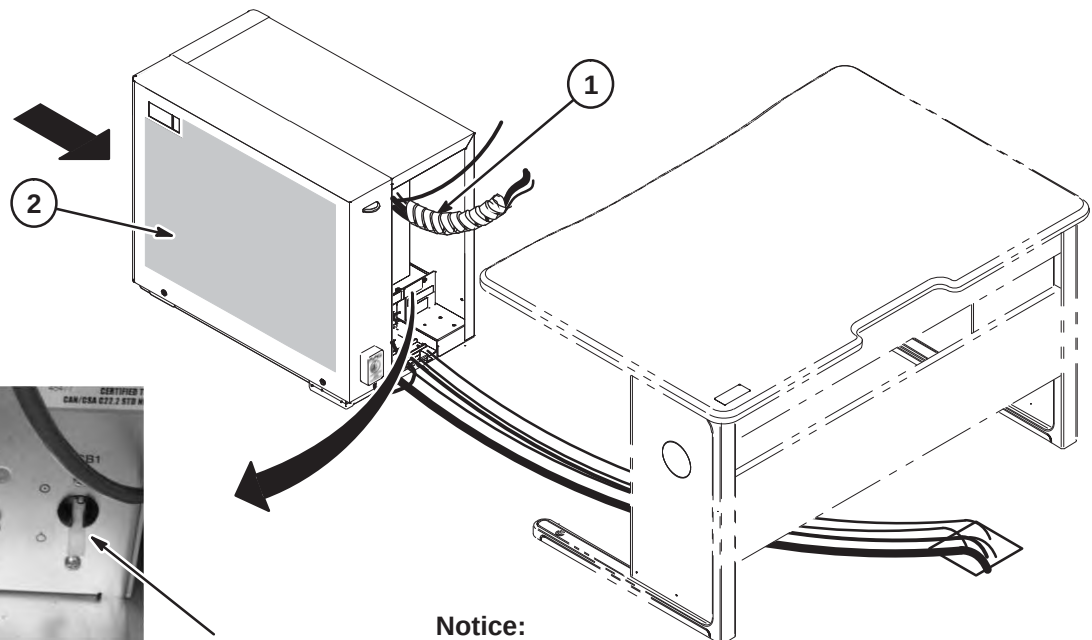
1. Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m).

Notice

Do Not include Modem power cable in Spiral Tube.

- Two power cords
- One video cable
- Keyboard cable
- SCIM cable
- Modem cable (if applicable)

2. Place cabinet in final position as determined by customer.



Notice:

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

5-11 POWER-ON INFORMATION

1. Remove LOTO from the PDU.

IMPORTANT

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the GOC AA and Modem power.

2. Restore power to the GOC, boot software, and ensure proper system operation.

NOTE: If necessary, reconnect the optional UPS accessory and ensure power is restored.

3. Perform the applicable Loading Host System Software procedure to load software and applications onto the operating system.

NOTE: The Load from Cold must be performed to properly transfer customer options

4. Update the elicense with the new Host ID.
5. Repackage the replaced computer for return
6. Perform a scan to ensure the system and applications are working properly.

SECTION 6 – SYSTEM CABINET UPGRADE

This section provides information on upgrading the existing HD System or RFS cabinet to HDx or HDxt. The upgrade procedures refer to parts being relocated, replaced, and removed for 1.5T and 3.0T systems that were either installed new for HD or upgraded to HD.

Refer to the applicable section for the site being upgraded:

- Section 6–1 – System cabinet upgrade instructions for 1.5T HD site
- Section 6–2 – RFS cabinet upgrade instructions for 1.5T HD site
- Section 6–3 – System cabinet upgrade instructions for 3.0T HD site
- Section 6–4 – RFS cabinet upgrade instructions for 3.0T HD site

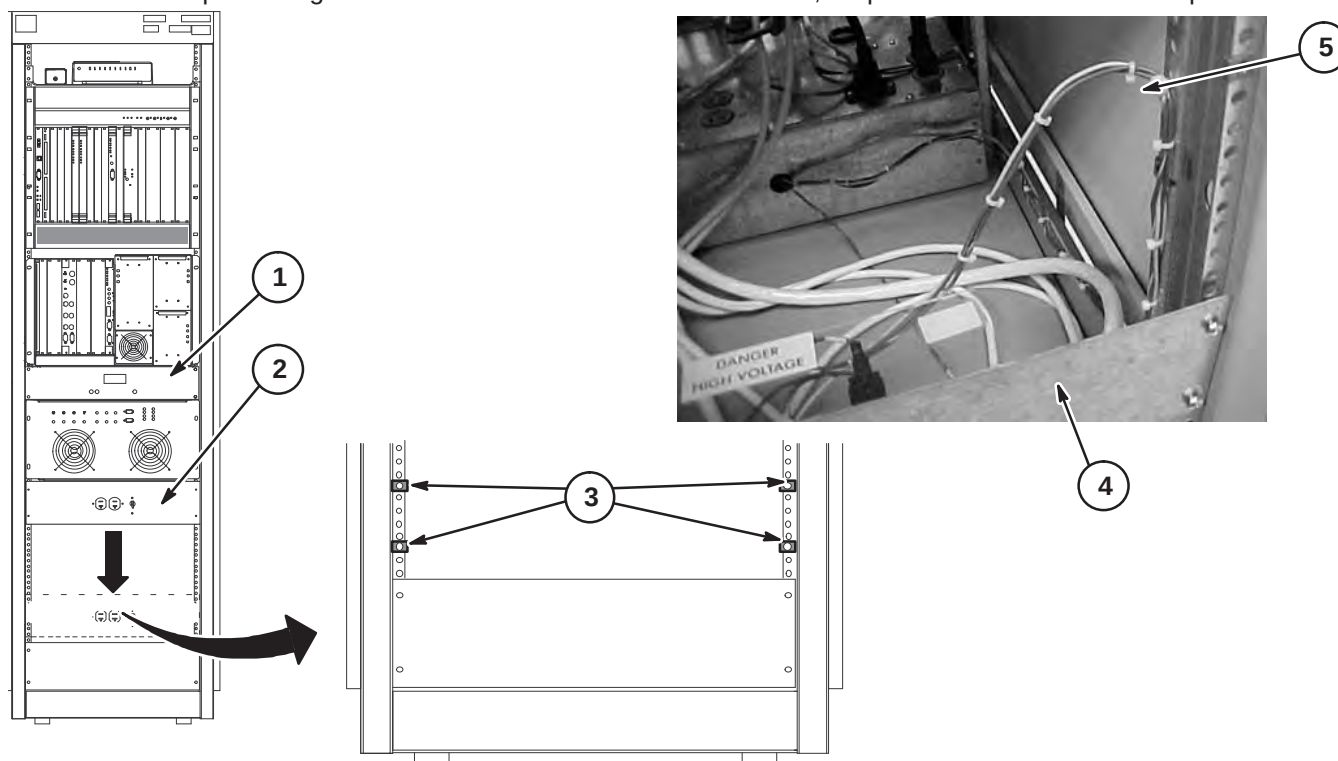
All de-installed modules and materials are to be returned for recycling.

6–1 SYSTEM CABINET UPGRADE INSTRUCTIONS FOR 1.5T HD SITE

The 1.5T HD System Cabinet upgrade relocates the interface panel at the back of the cabinet and installs the new VRE modules and cables.

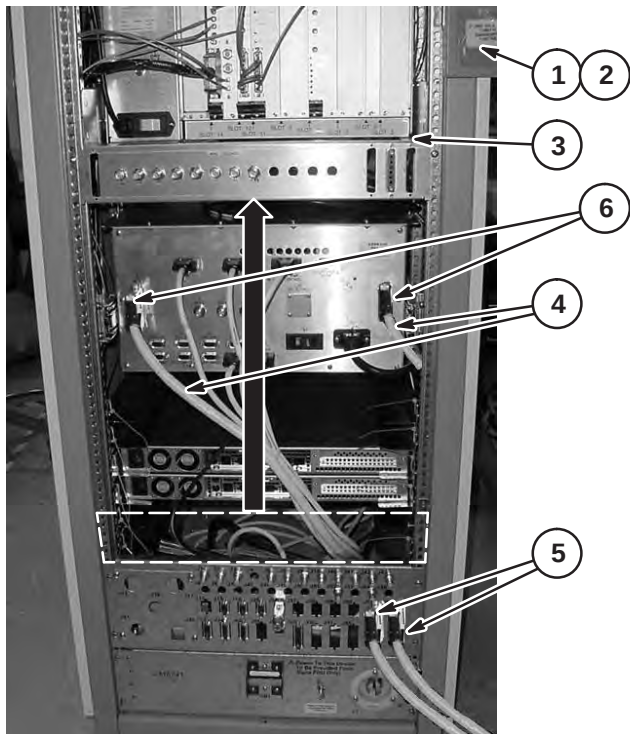
6–1–1 RELOCATE SERVICE POWER PANEL

1. If Cryogen Meter is part of cabinet, de-install and return for recycling.
2. Service Power Panel has to be relocated down to new location.
3. Install Clip Nuts at the 3rd and 8th hole above the blank panel on each vertical rail.
4. Fasten the Panel in place using the same mounting hardware. There will be a gap between Service Power Panel and the blank panel below.
5. Cut tie-wraps holding Service Power Panel wires to vertical rail, re-position cables and tie-wrap as shown.



6-1-2 UPGRADE BACK OF CABINET

1. De-install existing rear door and replace with new Rear Door (2394019-2).
2. Reconnect door ground wire.
3. Relocate Interface Panel to this location and align top of Panel to with bottom of RRF Chassis.
4. Disconnect and remove cables routed from:
 - Driver Module J5 to Interface Panel J61
 - Driver Module J10 to Interface Panel J66



Note:
Remainder of cables connected to Interface panels are not shown for clarity.

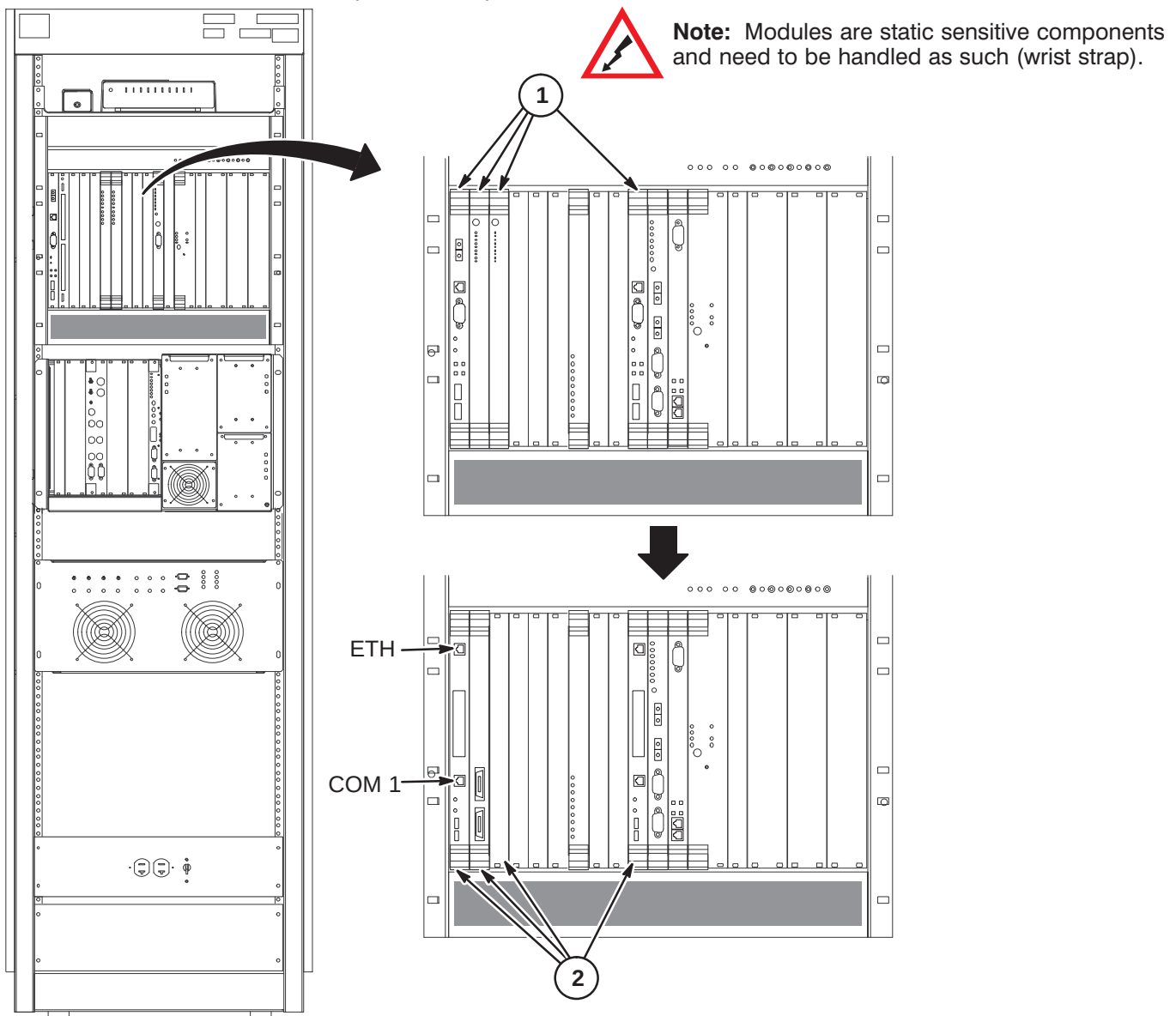
5. Disconnect Runs 1030 and 1251 from J61 and J66 and relabel per instructions on table below.

OLD LABEL	ONE END		OLD LABEL	OTHER END	
		NEW LABEL			NEW LABEL
RUN #1030 PP1-J78 2329624	RUN #1329 PP1-J78 5160765		RUN #1030 MR2-A11-J61 2329624	RUN #1329 MR2-DVR-J10 5160765	
RUN #1251 PP1-J203 5113186-11	RUN #1330 PP1-J203 5160765-4		RUN #1251 MR2-J66 5113186-11	RUN #1330 MR2-DVR-J5 5160765-4	

6. Route and connect relabeled Runs 1329 and 1330 directly to Driver Module and connect to J5 and J10 as marked.

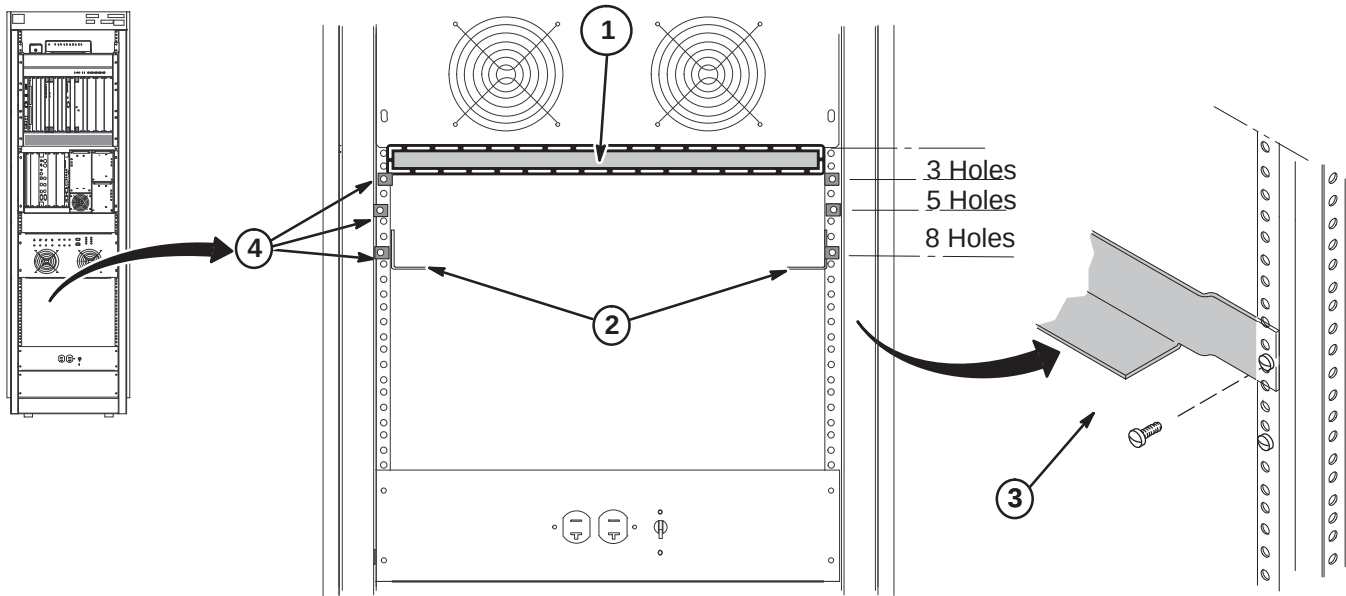
6-1-3 INSTALL NEW MGD CHASSIS MODULES

1. De-install and return for recycling the following boards:
 - Slot 1 – APS Board
 - Slot 2 & 3 – AP Boards
 - Slot 10 – AGP Board
2. Install the following new boards:
 - Slot 1 – APS Board (2294300-18)
 - Slot 2 – Infiniband Board (5139563)
 - Slot 3 – blank cover (2319152)
 - Slot 10 – AGP Board (2294300-18)



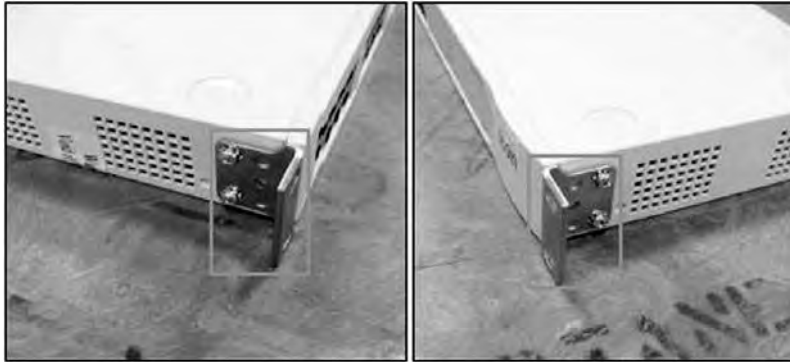
6-1-4 INSTALL VRE MODULE RAILS

1. Make sure that the clip nuts are located below two mounting holes on the vertical rail. This area between Driver Module and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.
2. Location of rails (5158758), one on each side, at the positions shown for the ICN Module.
3. Position rails behind vertical rails and attach with M5 Screws (2279742-2). Rails are attached at hole locations 8 of vertical rail.
4. For securing VRE modules – Install 3 Clip Nuts (2180698-2) on front of vertical rail on each side at locations indicated.

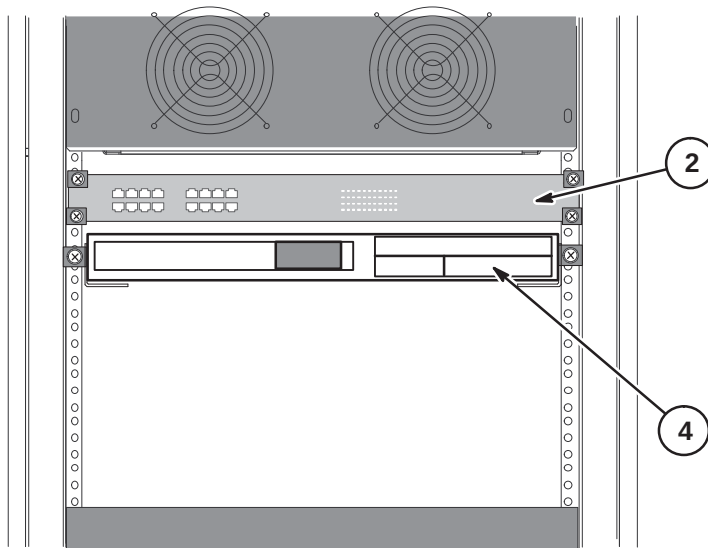


6-1-5 INSTALL VRE MODULES

1. Install brackets on front corners of Ethernet Switch as shown. Hardware kit comes with unit.



2. Install Ethernet Switch at this location using M5 Screws (2279742-2).
3. Install two (5306485) brackets on the ICN as shown. The brackets attach by snapping them into place.
4. Install ICN Module (5337894-3) on Slide Rail at the location shown. Tighten the PEM screws on the ICN brackets to secure.



On the left side, the bracket is installed downward and snapped into place.



On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.



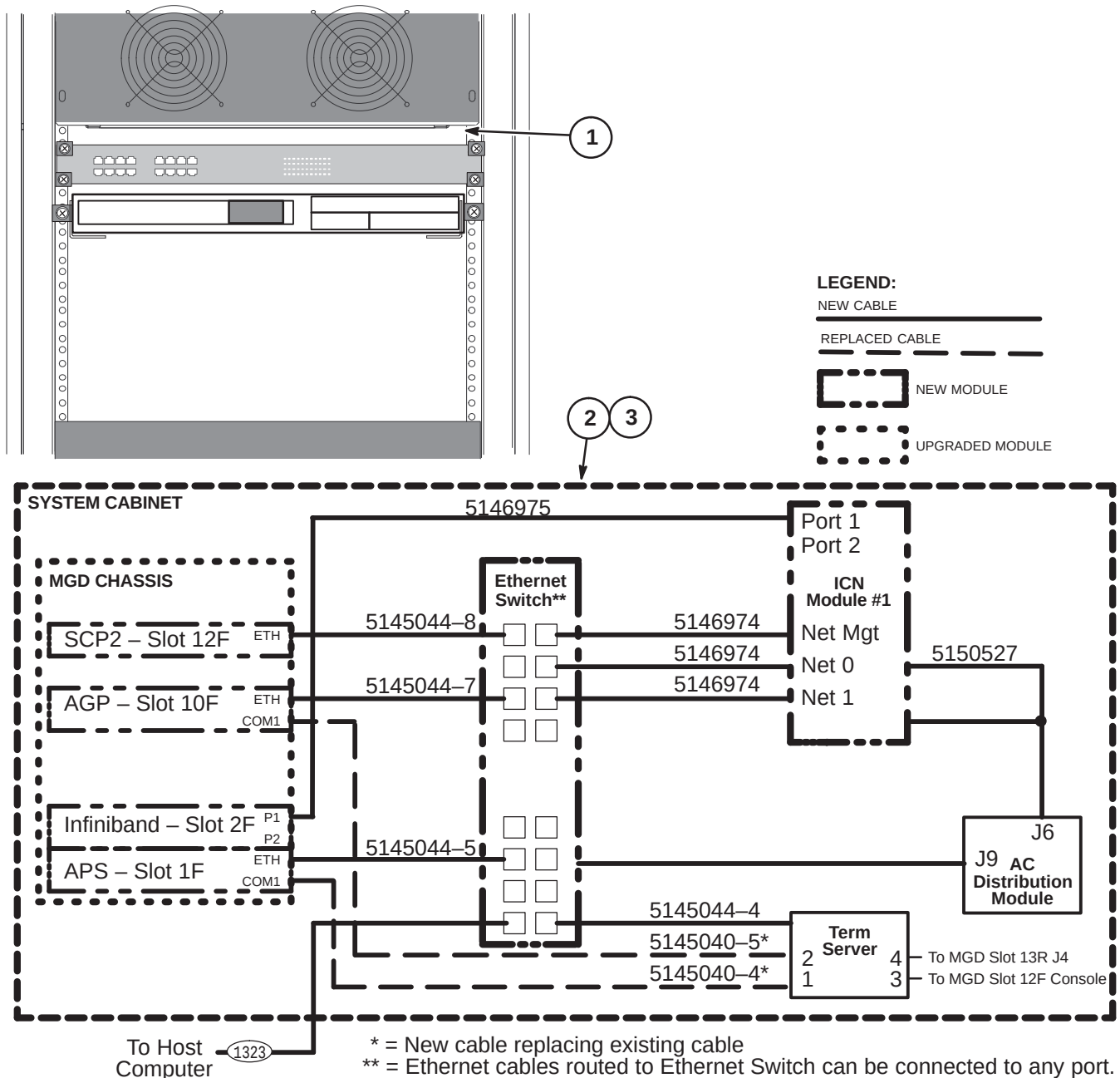
6-1-6 INSTALL VRE CABLES

Route and connect new VRE Cables as shown below.

1. All cables connecting between Ethernet Switch and ICN Module must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.
2. Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables. Make sure 5146975 is connected to P1 (Port 1) and not P2 on the back of the ICN.
3. Use power cable supplied with Ethernet Switch and connect to J9.

NOTE: Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.

4. Secure all modules in place with supplied fastening hardware.

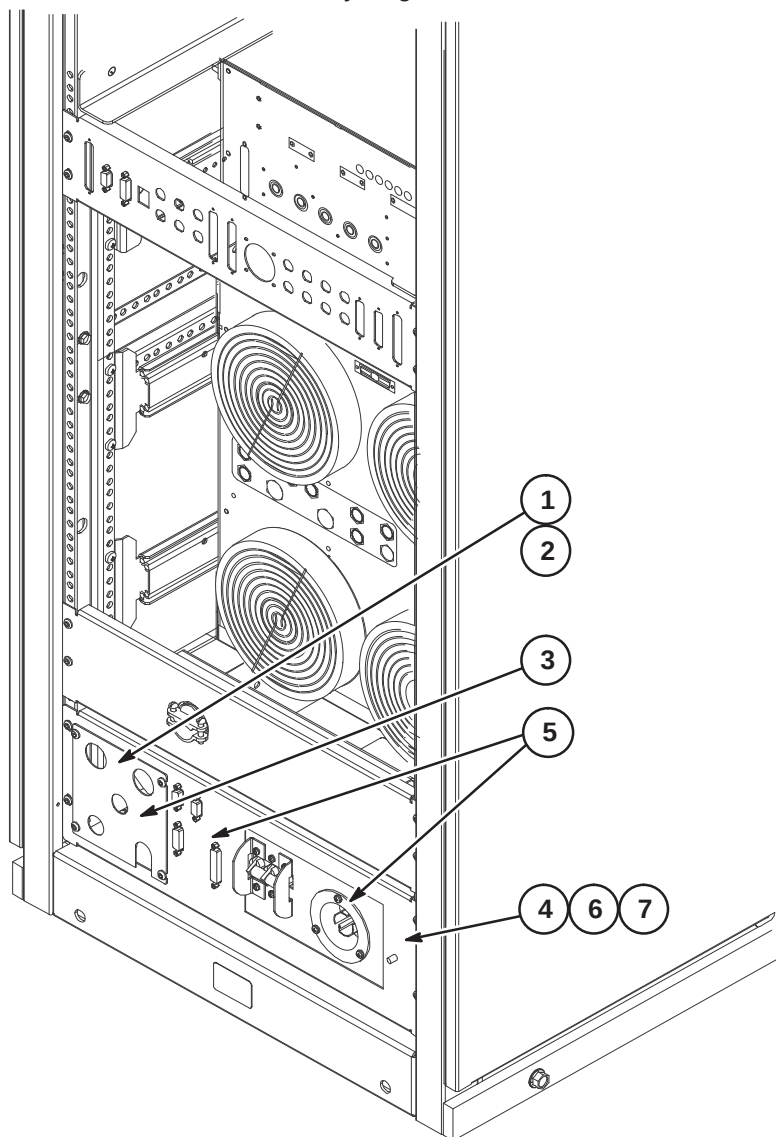


6-2 RF/SYSTEM CABINET UPGRADE INSTRUCTIONS FOR 1.5T HD SITE

The 1.5T RFS Cabinet upgrade replaces and relocates the interface panel at the back of the cabinet and installs the new VRE modules and cables.

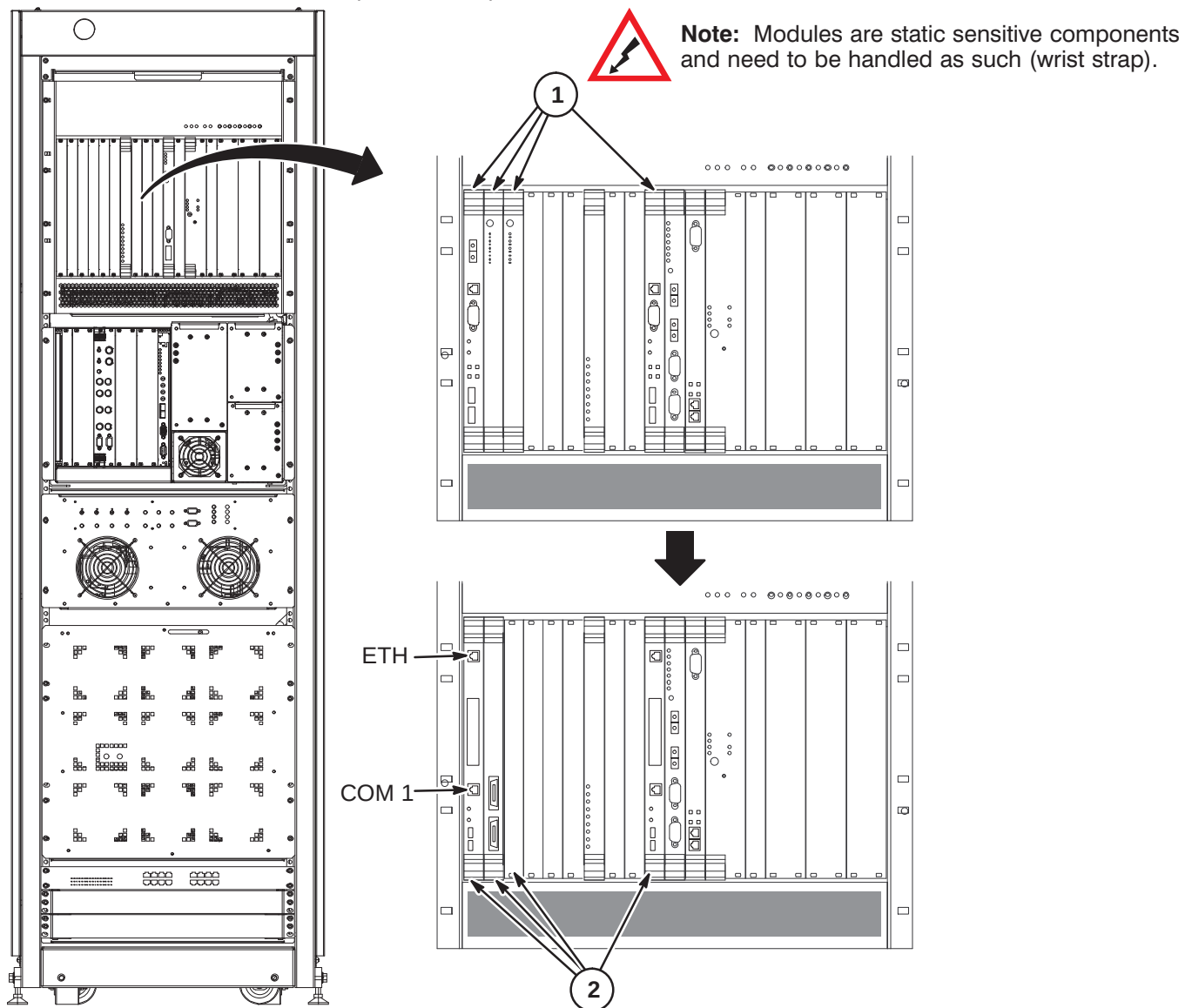
6-2-1 DE-INSTALL EXISTING POWER PANEL AT BACK OF CABINET

1. Disconnect all fiber optic leads connected to modules within cabinet.
2. Unfasten fiber optic connectors from mounting plate. Save Mounting Plate for reconnecting fiber optic conduits at new location.
3. Unfasten, remove, and save fiber optic mounting plate. Plate will be installed on new Interface Panel.
4. Remove ground cable and save nut. Remove screws on right side of Power Panel and loosen, but do not remove, screws on left side of panel.
5. Disconnect all power and data cables connected to Interface panel, both inside and outside of cabinet. Use care to capture data cable hardware.
6. Unfasten remaining screws securing Power Panel to cabinet and de-install panel.
7. Return Power Panel for recycling.



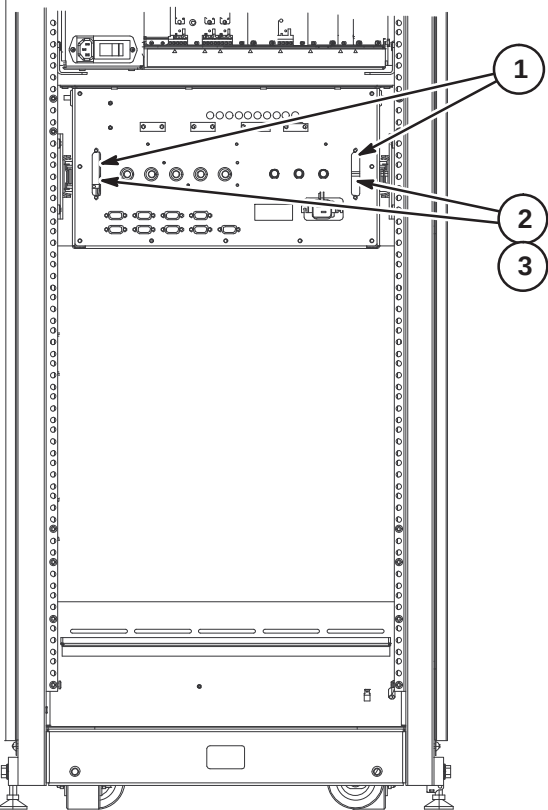
6-2-2 INSTALL NEW MGD CHASSIS MODULES

1. De-install and return for recycling the following boards:
 - Slot 1 – APS Board
 - Slot 2 & 3 – AP Boards
 - Slot 10 – AGP Board
2. Install the following new boards:
 - Slot 1 – APS Board (2294300-18)
 - Slot 2 – Infiniband Board (5139563)
 - Slot 3 – blank cover (2319152)
 - Slot 10 – AGP Board (2294300-18)



6-2-3 RELABEL AND SWAP DRIVER MODULE CABLES AT J5 AND J10

1. Disconnect Runs 1189 and 1190 from J5 and J10 on Driver Module.



Note:
Relabeled cables are swapped from original connections.

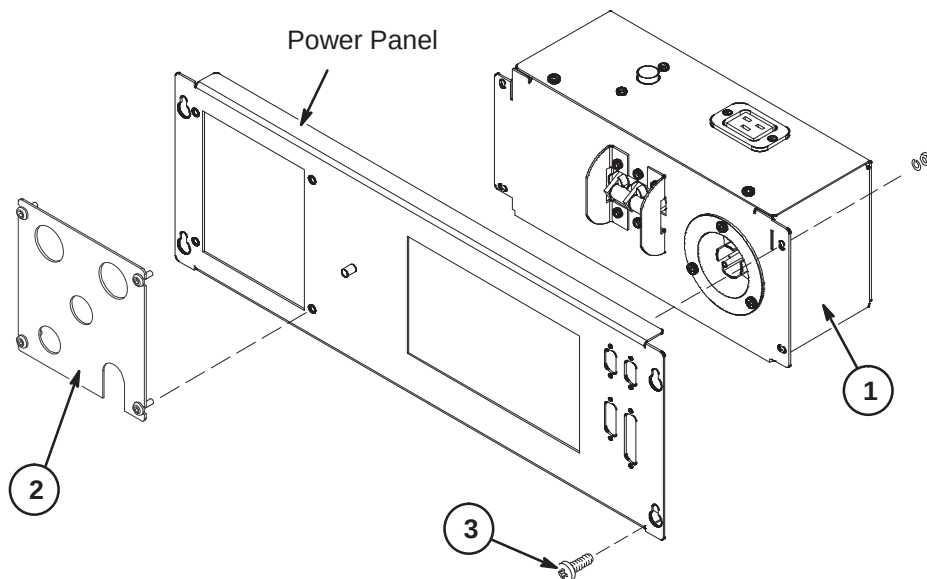
2. Relabel ends of cables with instructions provided in table below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1189 PP1-J78 5113186-2	RUN #1329 PP1-J78 5160765	RUN #1189 MR2-DVR-J5 5113186-2	RUN #1329 MR2-DVR-J10 5160765
RUN #1190 PP1-J203 5113186-5	RUN #1330 PP1-J203 5160765-4	RUN #1190 MR2-DVR-J10 5113186-5	RUN #1330 MR2-DVR-J5 5160765-4

3. Connect cable Run 1329 (J10) and Run 1330 (J5) to Driver Module per new label information.
Note: Cables are swapped from their original positions.

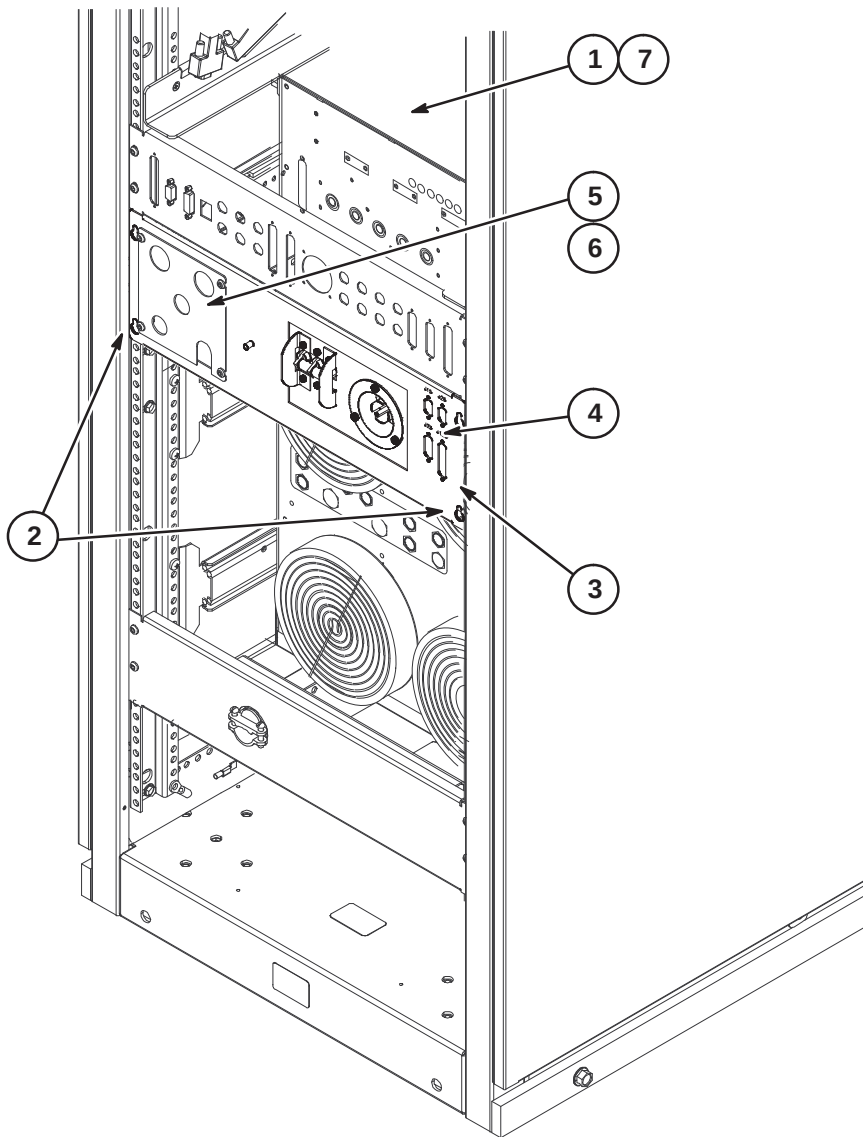
6-2-4 ASSEMBLE NEW POWER PANEL

1. Attach AC Distribution Box (5139347) to Power Panel (5147682) using four M4 Nuts (46-328425P1) and four M4 Lock Washers (46-328433P2).
2. Attach Fiber Optic Mounting Plate, transferred from de-installed Interface Panel, with existing hardware.
3. New Interface Panel Assembly will be attached to cabinet using four M5 x 12mm Flat Head Screws.



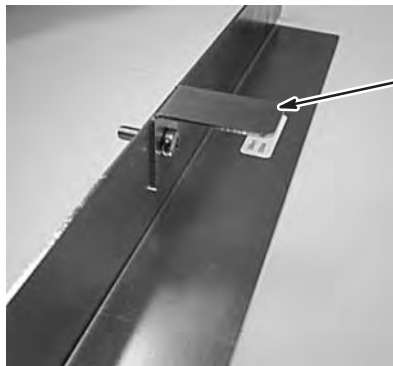
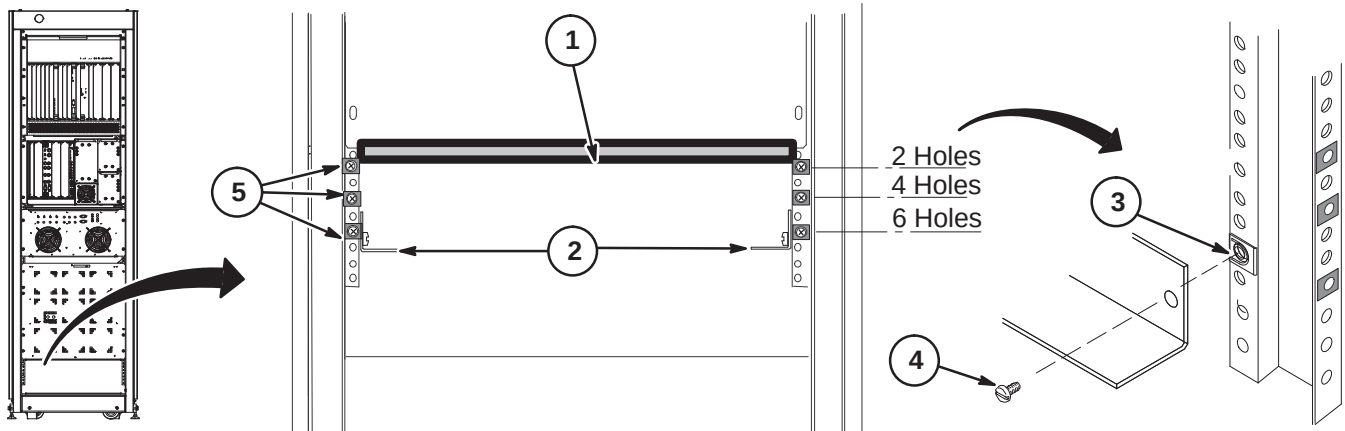
6-2-5 INSTALL NEW POWER PANEL AT BACK OF CABINET

1. Remove screws retaining Driver Module at front of cabinet and extend Driver Module from cabinet 2 to 3 inches (50 to 75mm).
2. Position four Nut Clips (2180698-2), two on each vertical rail, so that new Interface Panel Assembly can be mounted directly below existing I/O Panel.
3. Attach Interface Panel Assembly using four M5 x 12mm (46-312234P46) Flat Head Screws.
Note: If required, reroute Multicoil Driver cables to left and/or right side of cabinet to make room for new panel assembly.
4. Connect all previously disconnected power and data cables to Interface panel, both inside and outside of cabinet.
5. Re-attach fiber optic conduits, Runs 1034 and 1045, to fiber optic plate. Run 1083 has been deleted from this upgrade and is not re-installed.
6. Route and re-connect all fiber optic lead to modules within cabinet.
7. Reinstall Driver Module.



6-2-6 INSTALL VRE MODULE RAILS

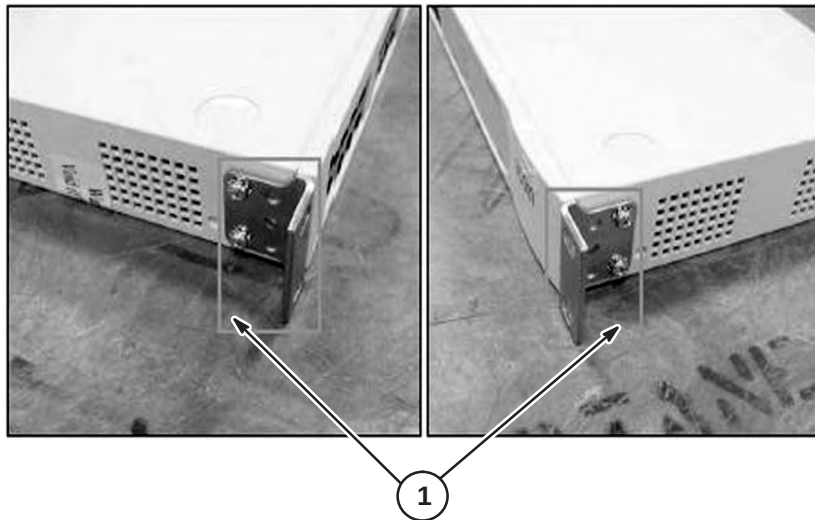
1. Make sure that the clip nuts are located below one mounting hole on the vertical rail. This area between RF chassis and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.
2. Location of Rails (5306484), one on each side.
3. Position four Nut Clips (2180698-2), one on each vertical rail, at front and rear of cabinet, at locations shown.
4. Attach Rails using M5 x 12mm (46-312234P46) Flat Head Screws. Rails are attached at hole locations 6 of vertical rail.
5. Install 3 Clip Nuts (2180698-2) on front surface of each front vertical rail at locations indicated.
6. Install one (5306487) L-brackets at the back of the cabinet on top rail only.



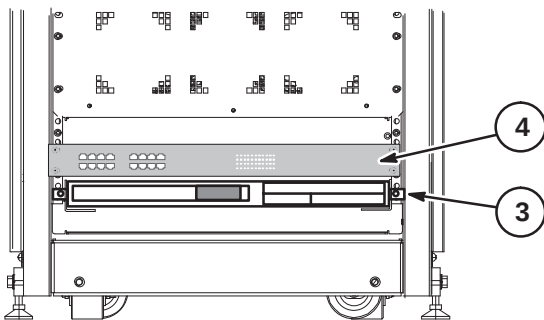
NOTE: Bracket and rail shown removed from cabinet for clarity.

6-2-7 INSTALL VRE MODULE

1. Install brackets on front corners of Ethernet switch as shown. Hardware kit comes with unit.



2. Install two (5306485) brackets on the ICN as shown.
3. Install ICN Module (5337894-3) on Top Rail at the location shown. Tighten PEM screws on the ICN brackets to secure.
4. Install Ethernet Switch at this location. Make sure power switch is turned on.



On the left side, the bracket is installed downward and snapped into place.



On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

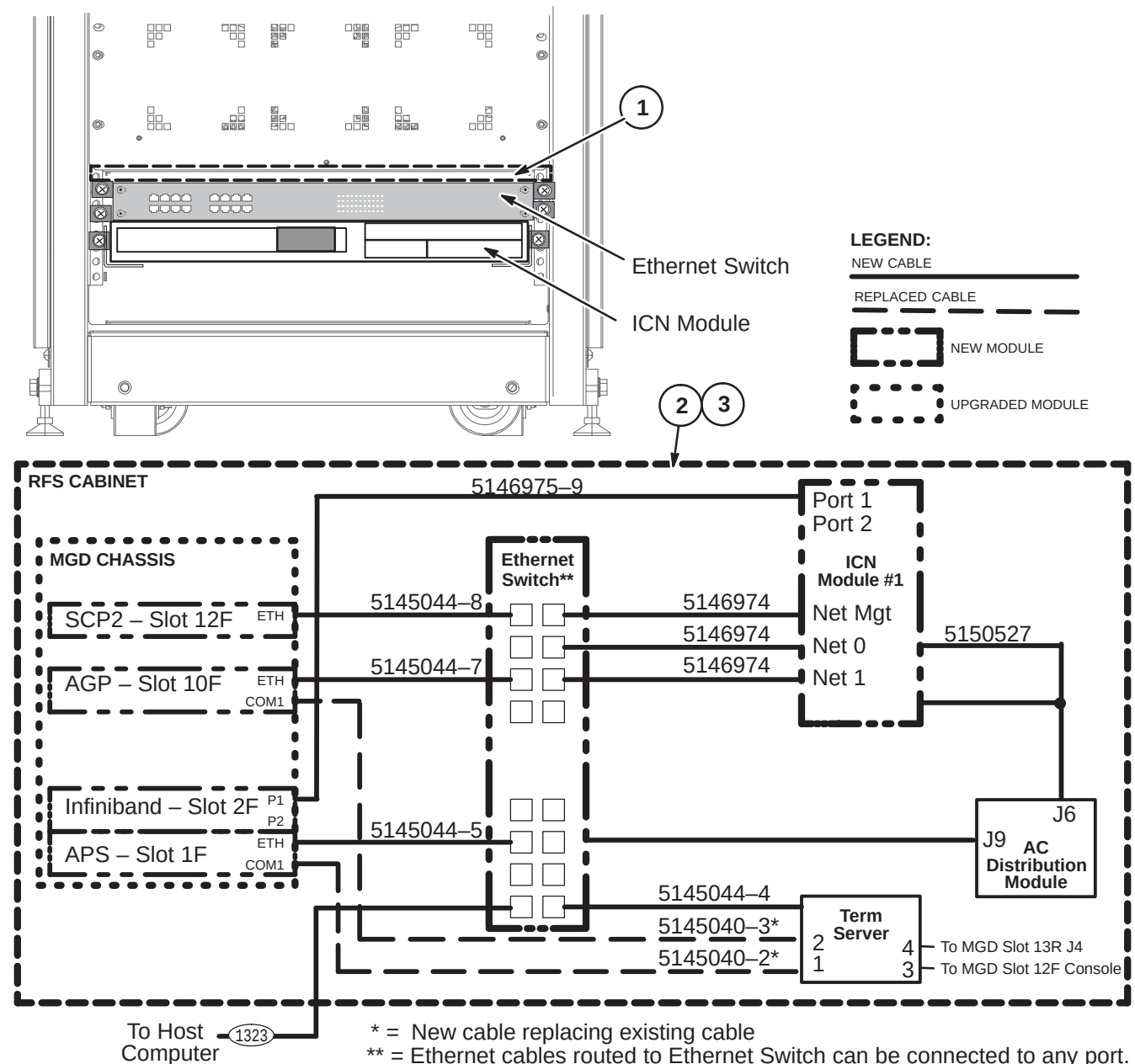


6-2-8 INSTALL VRE CABLES

1. All cables connecting between Ethernet Module and ICN Module must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.
2. Route and connect new VRE Cables as shown below. Cables 5145040-2 and 5145040-3 are replacing existing cables. Make sure 5146975 is connected to P1 (Port 1) and not P2 on the back of the ICN.
3. Use power cable supplied with Ethernet Switch and connect to J9.

NOTE: Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.

4. Secure all modules in place with supplied fastening hardware.

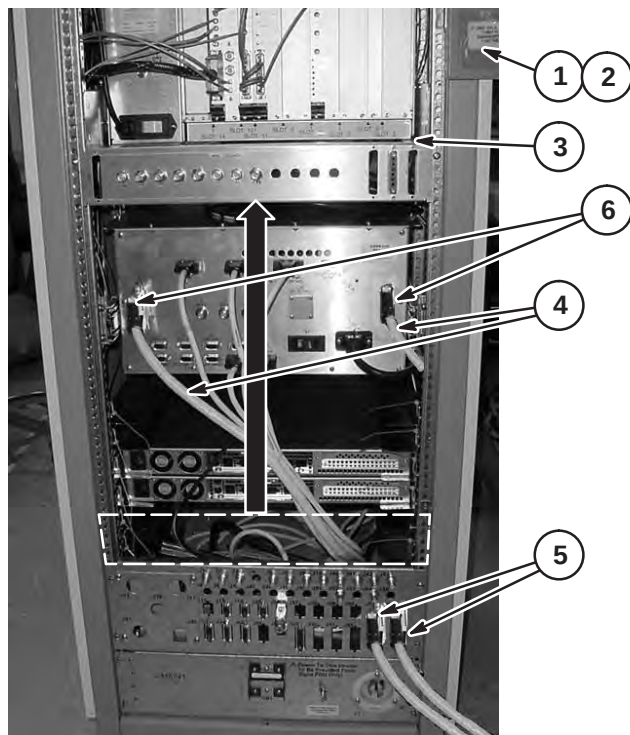


6-3 SYSTEM CABINET UPGRADE INSTRUCTIONS FOR 3.0T HD SITE

The 3.0T HD System Cabinet upgrade relocates the interface panel at the back of the cabinet and installs the new VRE modules and cables.

6-3-1 UPGRADE BACK OF CABINET

1. De-install existing rear door and replace with new Rear Door (2394019-2).
2. Reconnect door ground wire.
3. Relocate Interface Panel to this location and align top of Panel to with bottom of RRF Chassis.
4. Disconnect and remove cables routed from:
 - Driver Module J5 to Interface Panel J61
 - Driver Module J10 to Interface Panel J66



Note:
Remainder of cables connected to Interface panels are not shown for clarity.

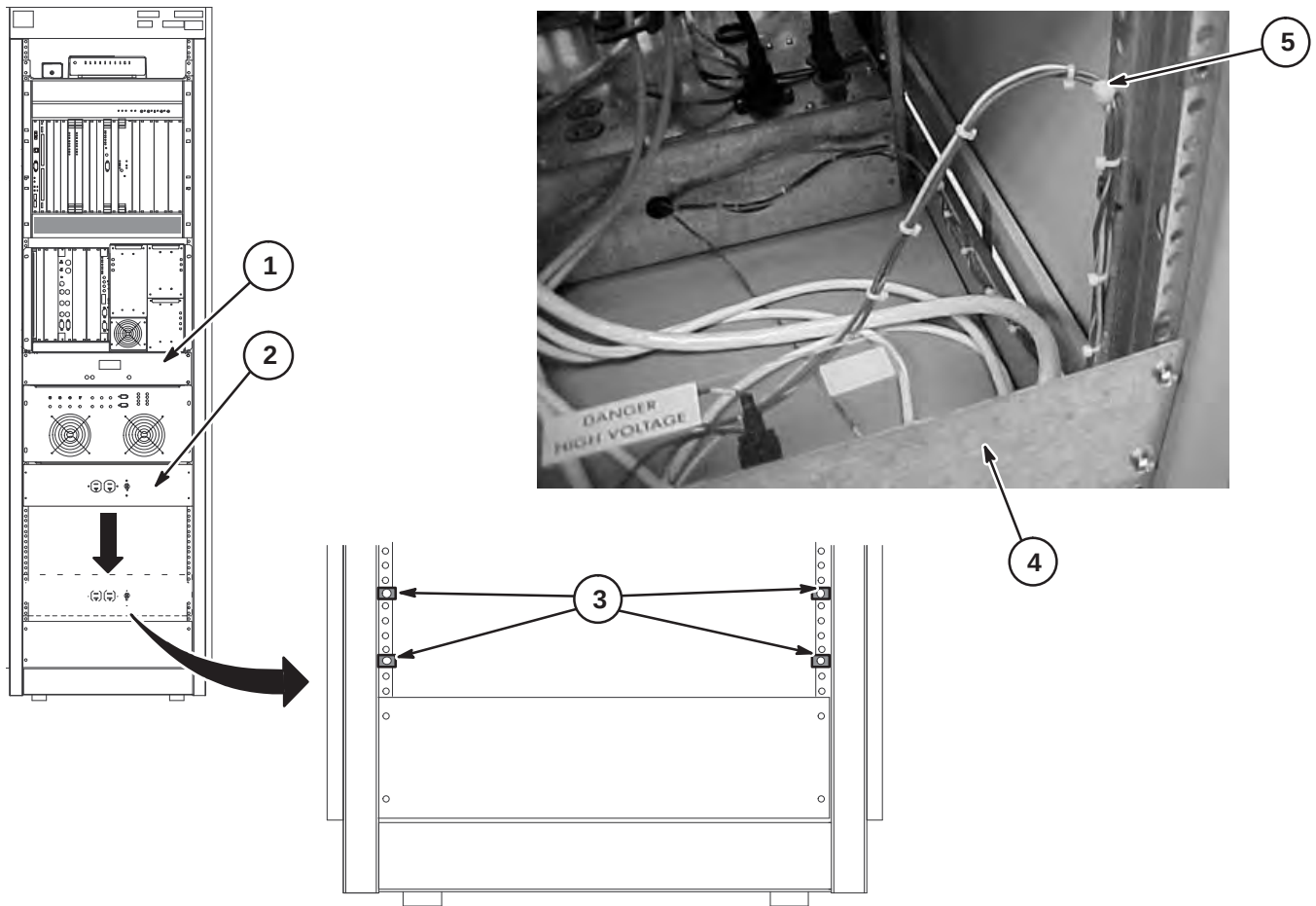
5. Disconnect Runs 1030 and 1251 from J61 and J66 and relabel per instructions on table below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1030 PP1-J78 2329624	RUN #1329 PP1-J78 5160765	RUN #1030 MR2-A11-J61 2329624	RUN #1329 MR2-DVR-J10 5160765
RUN #1251 PP1-J203 5113186-11	RUN #1330 PP1-J203 5160765-4	RUN #1251 MR2-J66 5113186-11	RUN #1330 MR2-DVR-J5 5160765-4

6. Route and connect relabeled Runs 1329 and 1330 directly to Driver Module and connect to J5 and J10 as marked.

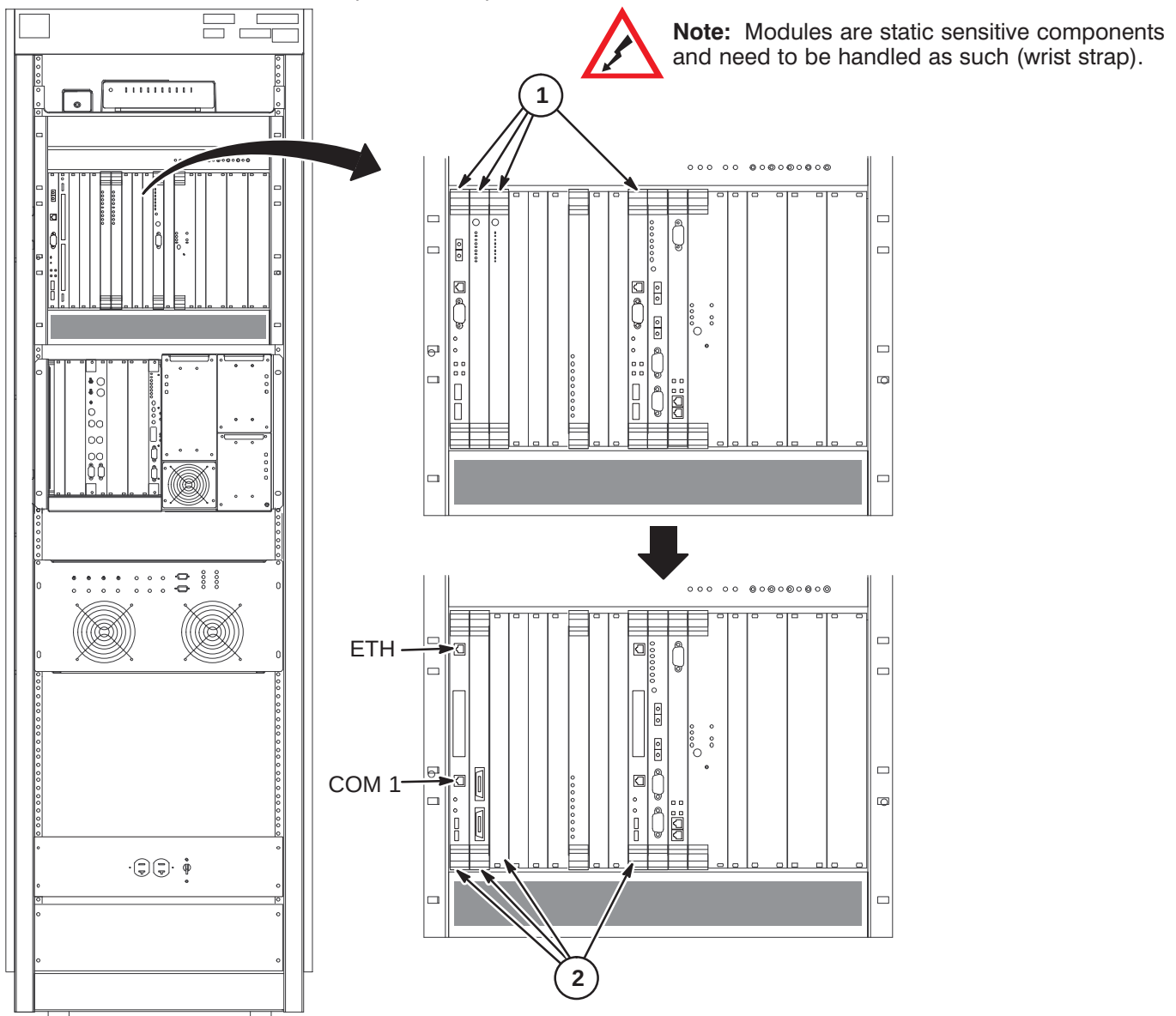
6-3-2 RELOCATE SERVICE POWER PANEL

1. If Cryogen Meter is part of cabinet, de-install and return for recycling.
2. Service Power Panel has to be relocated down to new location.
3. Install Clip Nuts at the 3rd and 8th hole above the blank panel on each vertical rail.
4. Fasten the Panel in place using the same mouting hardware. There will be a gap between Service Power Panel and the blank panel below.
5. Cut tie-wraps holding Service Power Panel wires to vertical rail, re-position cables and tie-wrap as shown.



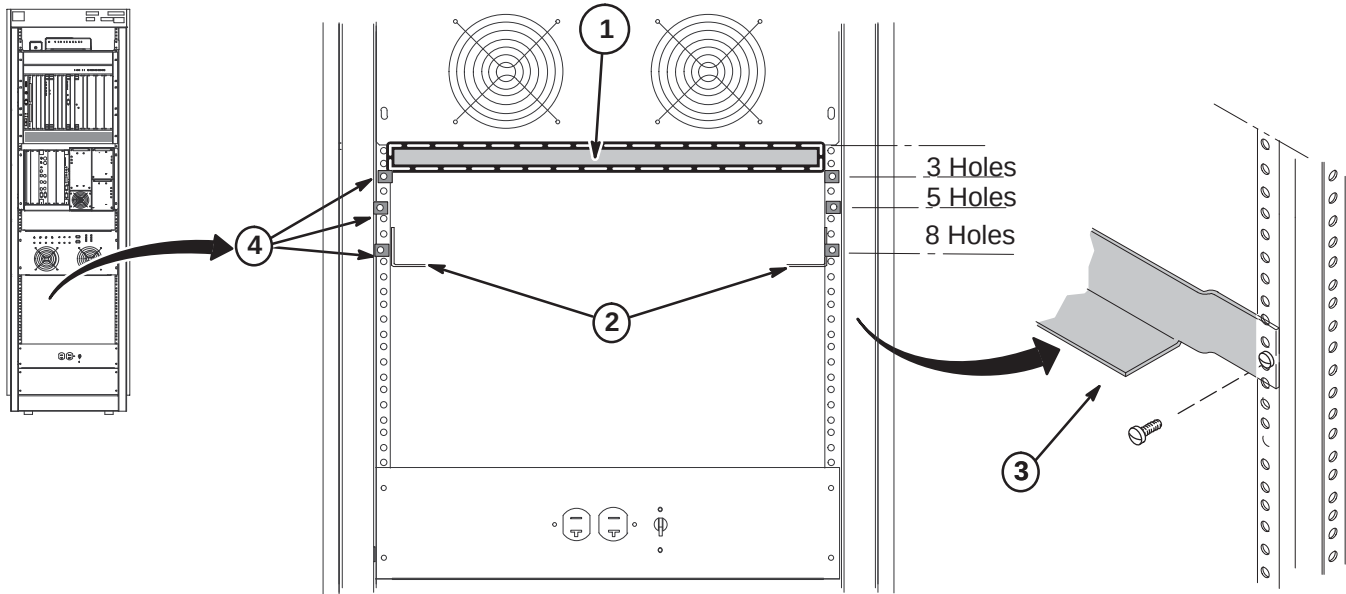
6-3-3 INSTALL NEW MGD CHASSIS MODULES

1. De-install and return for recycling the following boards:
 - Slot 1 – APS Board
 - Slot 2 & 3 – AP Boards
 - Slot 10 – AGP Board
2. Install the following new boards:
 - Slot 1 – APS Board (2294300-18)
 - Slot 2 – Infiniband Board (5139563)
 - Slot 3 – blank cover (2319152)
 - Slot 10 – AGP Board (2294300-18)



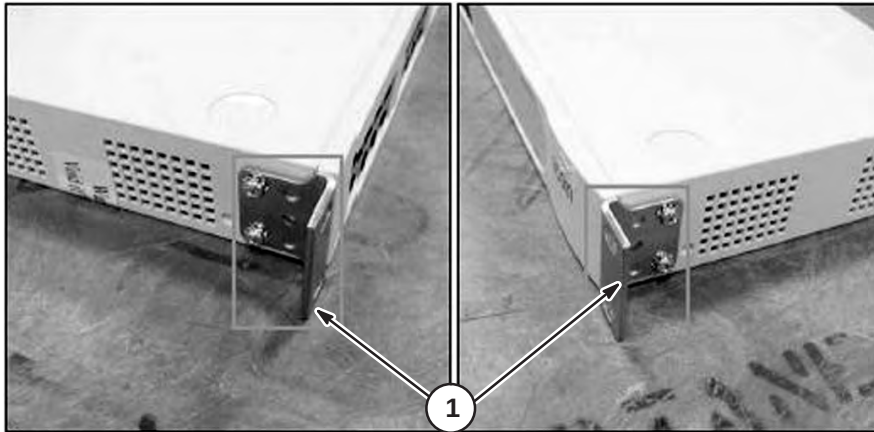
6-3-4 INSTALL VRE MODULE RAILS

1. Make sure that the clip nuts are located below two mounting holes on the vertical rail. This area between Driver Module and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.
2. Location of rails (5158758), one on each side, at the positions shown for the ICN Module.
3. Position rails behind vertical rails and attach with M5 Screws (2279742-2). Rails are attached at hole location 8 of vertical rail.
4. For securing VRE modules – Install 3 Clip Nuts (2180698-2), on front of vertical rail on each side at locations indicated.

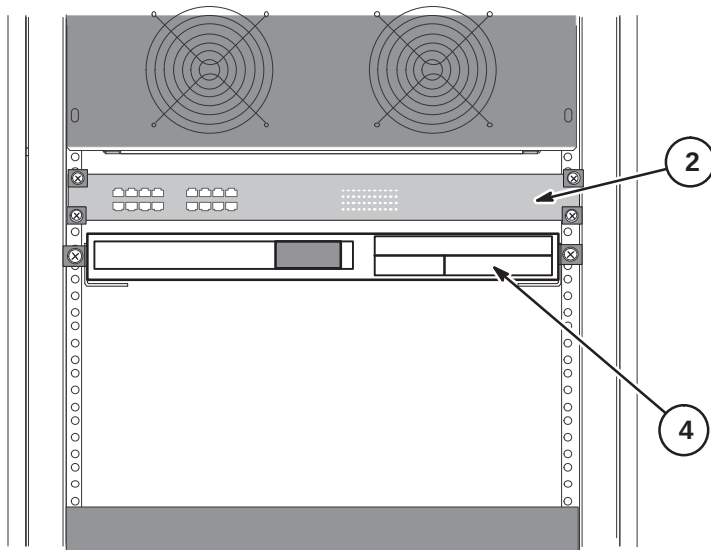


6-3-5 INSTALL VRE MODULES

1. Install brackets on front corners of Ethernet Switch as shown. Hardware kit comes with unit.



2. Install Ethernet Switch at this location using Clip Nuts and M5 Screws (2279742-2).
3. Install two (5306485) brackets on the ICN as shown.
4. Install ICN Module (5337893-3) on Rail at the location.



On the left side, the bracket is installed downward and snapped into place.

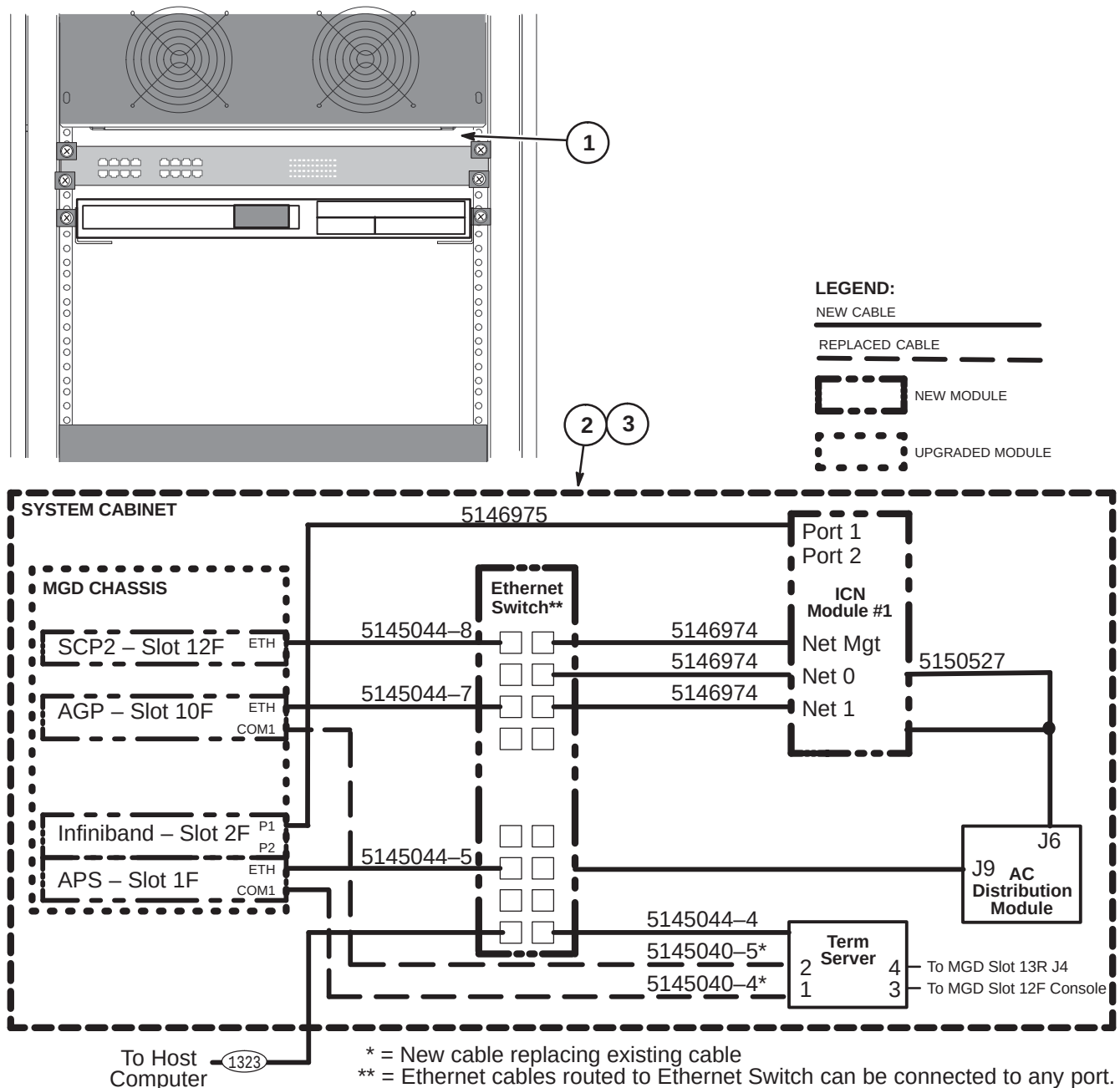


On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.



6-3-6 INSTALL VRE CABLES

1. All cables connecting between Ethernet Module and ICN Modules must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.
 2. Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables. Make sure 5146975 is connected to P1 (Port 1) and not P2 on the back of the ICN.
 3. Use power cable supplied with Ethernet Switch (not shown) and connect to J9.
- NOTE:** Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.
4. Secure all modules in place with supplied fastening hardware.

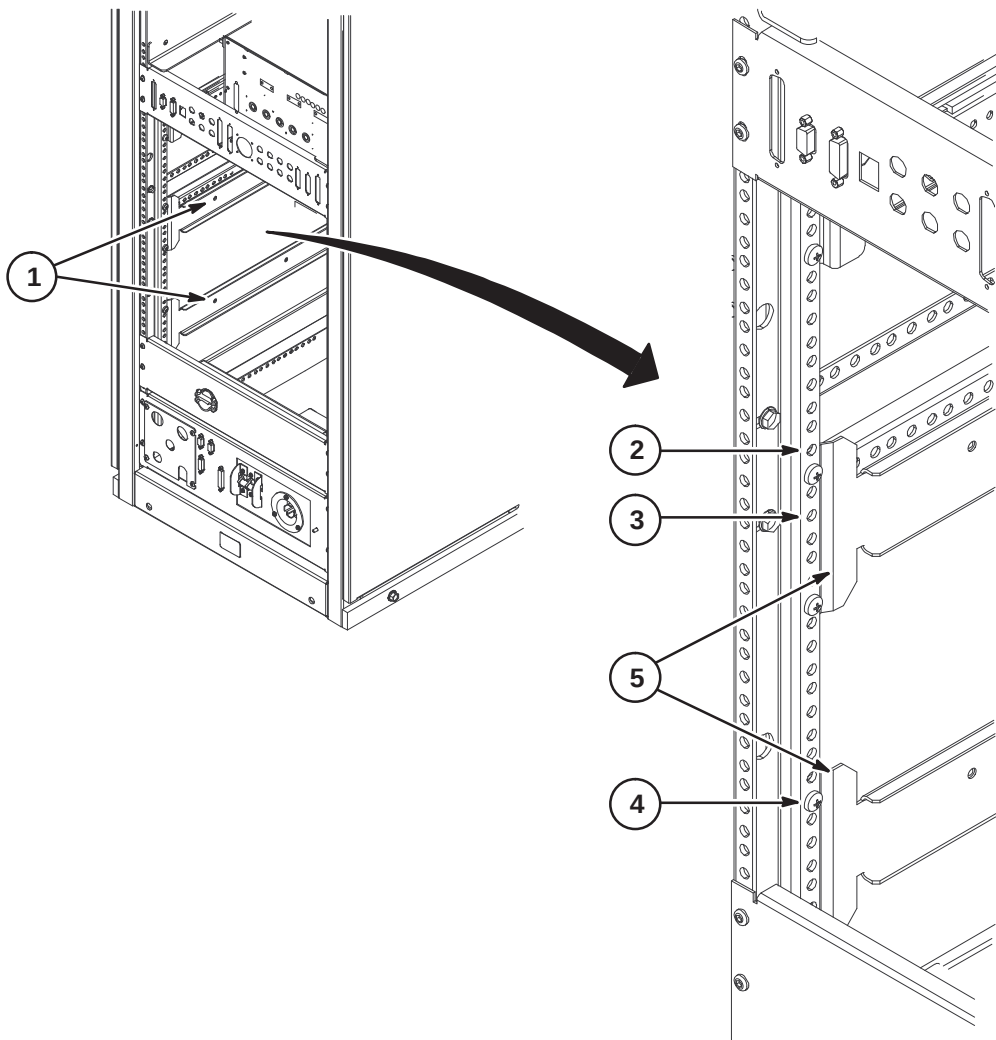


6-4 RF/SYSTEM CABINET UPGRADE INSTRUCTIONS FOR 3.0T HD SITE

The 3.0T RFS Cabinet upgrade removes rails inside of the cabinet and installs the new VRE modules and cables.

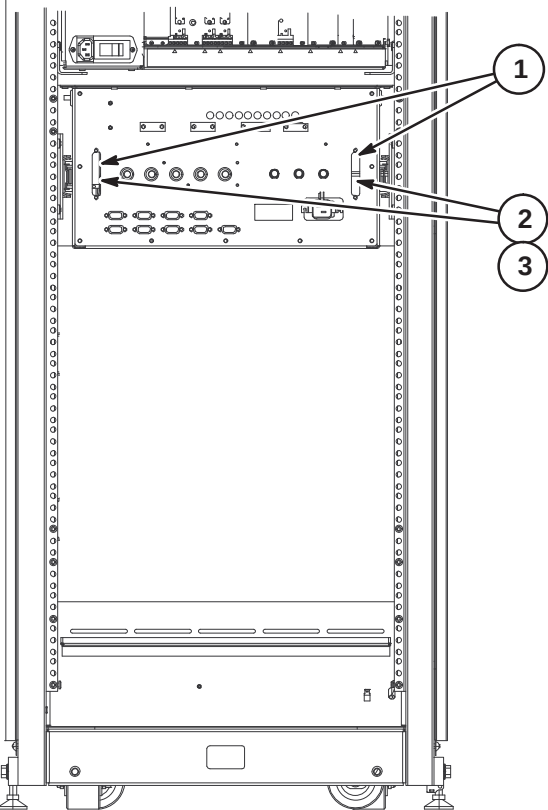
6-4-1 DE-INSTALL LOWER RAILS INSIDE OF CABINET

1. The two sets of rails on the left and right side of the cabinet have to be removed for the upgrade. **Before removing rails, the hole locations for the new VRE brackets have to be located and marked.**
2. Mark the hole on front and back vertical channels directly above the existing screw on the rail to be removed.
3. Mark the third hole down from the above marked hole on front and back vertical channels.
4. Mark the hole of the existing top screw of the lower rail on front and back vertical channels.
5. After all hole locations have been marked, remove the two sets of existing rails.



6-4-2 RELABEL AND SWAP DRIVER MODULE CABLES AT J5 AND J10

1. Disconnect Runs 1189 and 1190 from J5 and J10 on Driver Module.



Note:
Relabeled cables are swapped from original connections.

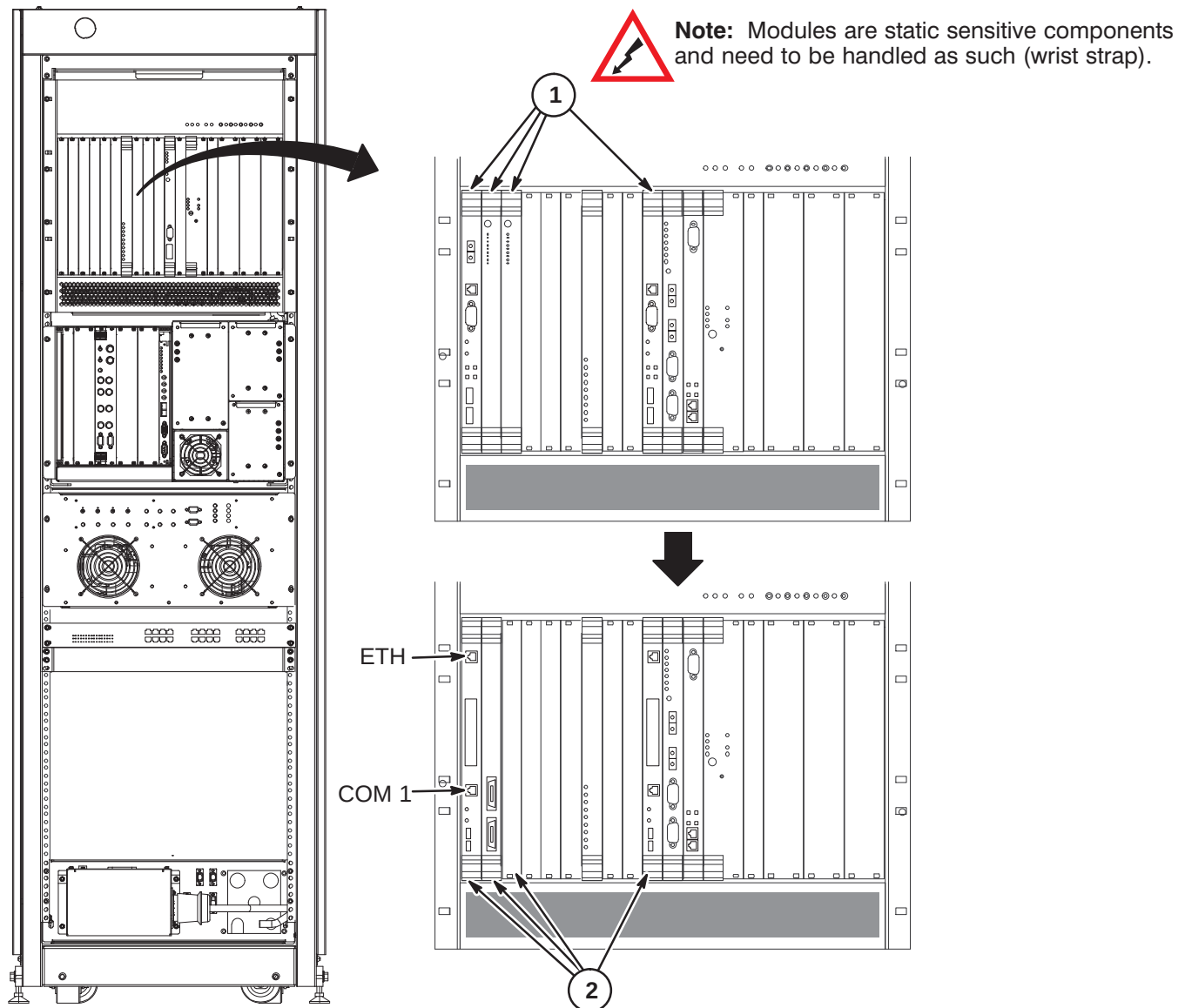
2. Relabel ends of cables with instructions provided in table below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1189 PP1-J78 5113186-2	RUN #1329 PP1-J78 5160765	RUN #1189 MR2-DVR-J5 5113186-2	RUN #1329 MR2-DVR-J10 5160765
RUN #1190 PP1-J203 5113186-5	RUN #1330 PP1-J203 5160765-4	RUN #1190 MR2-DVR-J10 5113186-5	RUN #1330 MR2-DVR-J5 5160765-4

3. Connect cable Run 1329 (J10) and Run 1330 (J5) to Driver Module per new label information.
Note: Cables are swapped from their original positions.

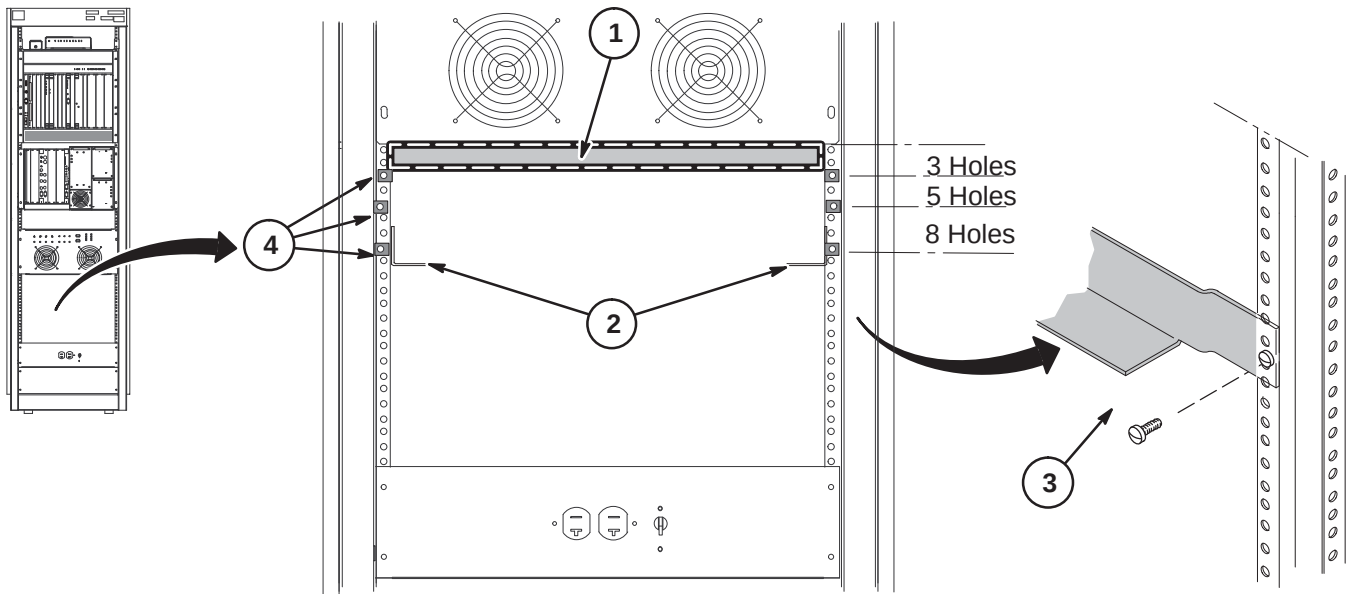
6-4-3 INSTALL NEW MGD CHASSIS MODULES

1. De-install and return for recycling the following boards:
 - Slot 1 – APS Board
 - Slot 2 & 3 – AP Boards
 - Slot 10 – AGP Board
2. Install the following new boards:
 - Slot 1 – APS Board (2294300-18)
 - Slot 2 – Infiniband Board (5139563)
 - Slot 3 – blank cover (2319152)
 - Slot 10 – AGP Board (2294300-18)



6-4-4 INSTALL VRE MODULE RAILS

1. Make sure that the clip nuts are located below two mounting holes on the vertical rail. This area between Driver Module and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.
2. Location of rails (5306484), one on each side, at the positions shown for the ICN Modules.
3. Position rails behind vertical rails and attach with M5 Screws (2279742-2). Rails are attached at hole location 8 of vertical rail.
4. For securing VRE modules – Install 3 Clip Nuts (2180698-2) on front of vertical rail on each side at locations indicated.
5. Install two (5306487) L-brackets at the back of the cabinet on top rail only.



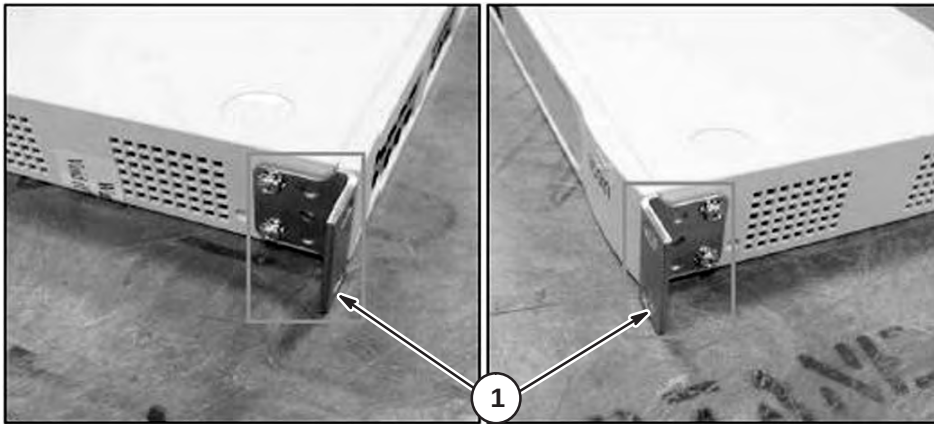
NOTE: The lower rail on each side is installed now for future installation of additionVRE option



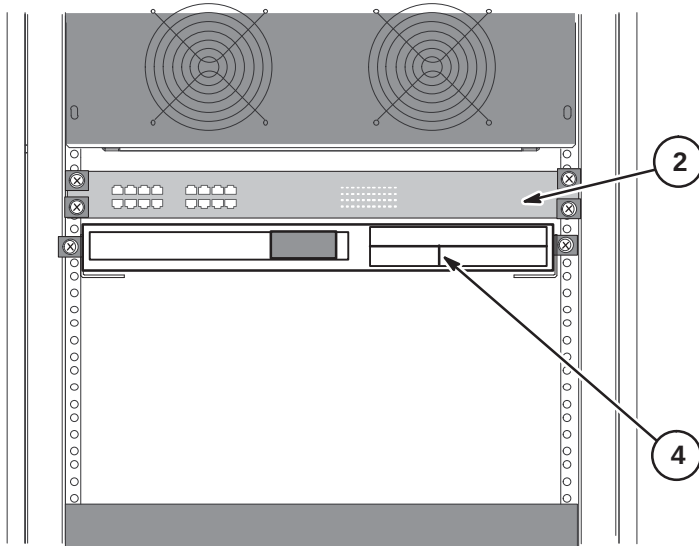
NOTE: Bracket and rail shown removed from cabinet for clarity.

6-4-5 INSTALL VRE MODULES

1. Install brackets on front corners of Ethernet Switch as shown. Hardware kit comes with unit.



2. Install Ethernet Switch at this location using M5 Screws (2279742-2).
3. Install two (5306485) brackets on the ICN as shown.
4. Install ICN Module (53377894-3) on Top Rail at the location shown.



On the left side, the bracket is installed downward and snapped into place.



On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

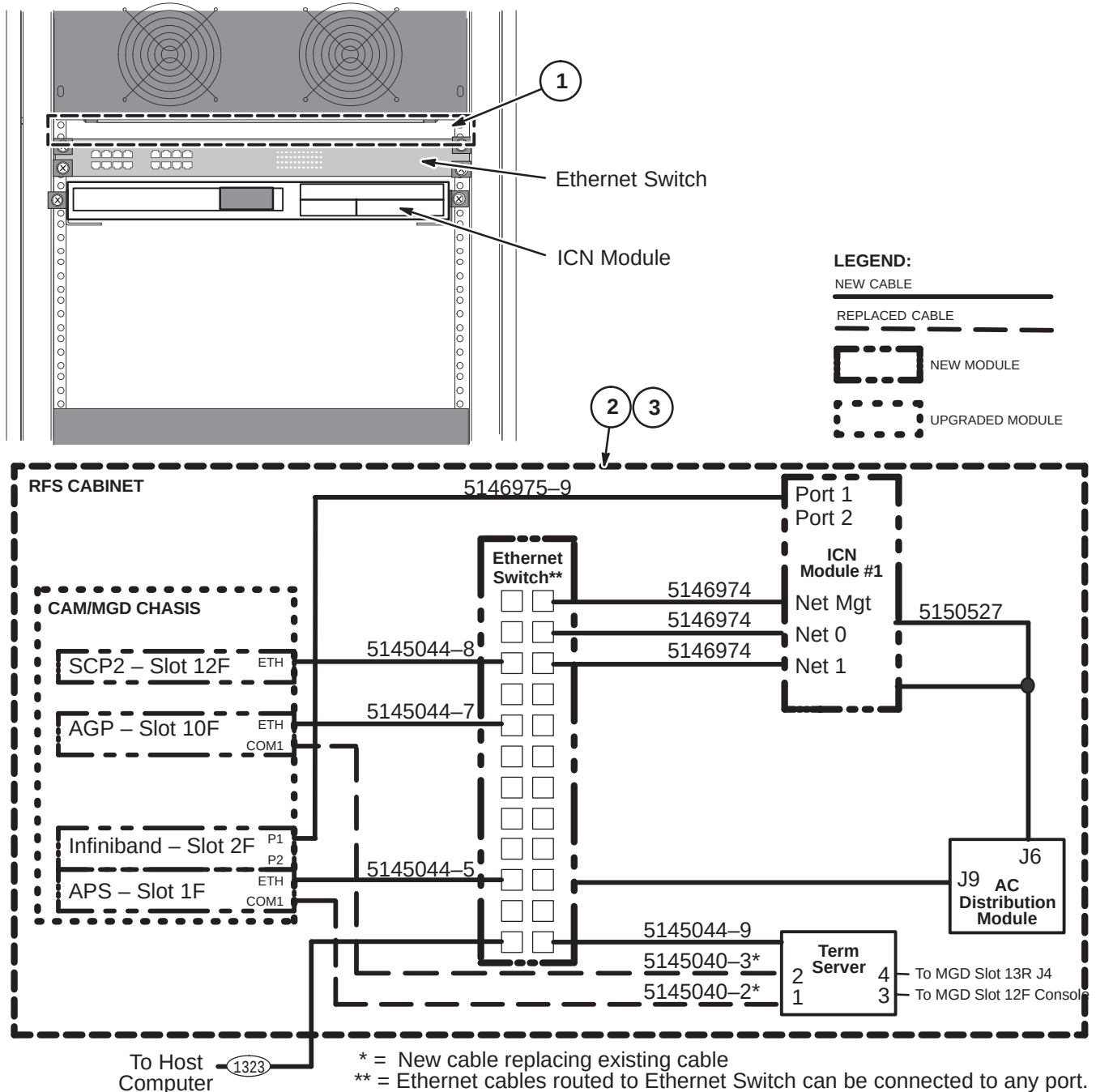


6-4-6 INSTALL VRE CABLES

1. All cables connecting between Ethernet Module and ICN Modules must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.
2. Route and connect new VRE Cables as shown below. Cables 5145040-2 and 5145040-3 are replacing existing cables. Make sure 5146975-9 is connected to P1 (Port 1) and not P2 on the back of the ICN.
3. Use power cable supplied with Ethernet Switch and connect to J9.

NOTE: Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.

4. Secure all modules in place with supplied fastening hardware.



SECTION 7 – COMPLETION

7-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 7-2, 7-3, and 7-4.

7-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 7-3. Recycling Operation will route them to appropriate location.

7-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

7-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers (If required)
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

7-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.